Probability and Stochastic Processes

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Chapter 2. Random Variable

- Section 2.1 The Definition of a Random Variable
- Section 2.2 The Distribution Function of a Random Variable
- Section 2.3 Mathematical Expectation and Variance
- Section 2.4 Discrete Random Variables
- Section 2.5 Continuous Random Variables

Section 2.1 The Definition of a Random Variable

The elements of a sample space may take diverse forms: real numbers, brands of components, colors, "good" or "defective" and so on.

In this chapter we transform all the elementary outcomes into numerical values, by means of random variables.

Two basic methods to study random variables. One is **distribution function**, which can describe random variable comprehensively. Another one is **numerical characteristic**, which gives us the information about the expectation, the derivation of a random variable.

The Definition of a Random Variable

In mathematics and empirical science, quantification is the act of counting and measuring that maps human sense observations and experiences into members of some set of numbers. Quantification in this sense is fundamental to the scientific method.

A random variable is a real-valued function defined on a sample space. It is a tool to quantify random events, to facilitate the use of mathematical expressions and to focus only on the outcomes of interest. It can help us understand random phenomenon from some concrete probabilities to the study of distribution, which can describe the random phenomenon comprehensively.

The Definition of a Random Variable

Definition

*A **random variable** is a function that assigns a real number to each outcome in the sample space.*

In a particular experiment a random variable X is a function that assigns a real number $X(\omega)$ to each $\omega \in \Omega$. The abbreviation for random variable $X(\omega)$ is $r.v.X$. We always use upper case letters (such as X, Y, T, U, V and so on) to denote random variables and lower case letters (x, y, z, etc.) to denote their possible numerical values.

The Definition of a Random Variable

Example

In the experiment of tossing a coin, we could get the outcome "Head" or "Tail". Let "Head" $= 1$ and "Tail" $= 0$. Then we can get a random variable "X" defined on $\Omega = \{\text{Head, Tail}\}$:

$$\begin{array}{ccc} \text{Random} & \text{Possible} & \text{Random} \\ \text{Variable} & \text{Values} & \text{Events} \\ \downarrow & \downarrow & \downarrow \end{array}$$

$$X = X(\omega) = \begin{cases} 1, & \omega_1 = \text{Head}, \\ 0, & \omega_2 = \text{Tail}. \end{cases}$$

If we toss n coins, let Y be the total number of heads shown by the n coins. Clearly, Y is a random variable defined on $\Omega = \{0, 1, 2, \cdots, n\}$.

The Definition of a Random Variable

Example

Suppose that our experiment consists of tossing three fair coins. Let X denote the number of heads appearing. Then X is a random variable defined on

$$\Omega = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$$

and it takes on one of the values 0, 1, 2 and 3. That is,

$$X(TTT) = 0, \qquad X(TTH) = X(THT) = X(HTT) = 1,$$

$$X(THH) = X(HTH) = X(HHT) = 2, \quad X(HHH) = 3.$$

What we should notice is it is possible to assign a same real number to different outcomes in sample space for some random variables.

The Definition of a Random Variable

Example

Consider the random experiment that consists in observing the time T that a person must wait in line to use an automatic teller machine. Then T is a random variable defined on $\Omega = [0, +\infty)$ and $T(t) = t$, $0 \leqslant t < \infty$.

Example

Let (x) denote a life aged x, where $x \geqslant 0$. The death of (x) can occur at any age greater than x, and we model the future lifetime of (x) by T_x. This means that $x + T_x$ represents the age-at-death random variable for (x). Then T_x is a random variable defined on $\Omega = [0, L - x)$, where L is the limiting age.

The Definition of a Random Variable

Observing the above examples, we can find the possible values of some random variables are finite, some are countably infinite and some are uncountably infinite. According to this point, we could divide the random variables into different categories.

Definition

A random variable is said to be of ***discrete type*** *if the number of different values it can take is finite or countably infinite.*

Question

Can we call a random variable continuous if it may take an uncountably infinite number of values? In most cases it is, but sometimes it is not.

Section 2.2 The Distribution Function of a Random Variable

- 2.2.1 The Definition and Properties of Distribution Function
- 2.2.2 The Distribution Function of Function of a Random Variable

The Definition and Properties of Distribution Function

Example

In the die experiment, we assign to the six outcomes ω_i the numbers $X(\omega_i) = 10i$. Thus

$$X(\omega_1) = 10,\ X(\omega_2) = 20,\ \cdots,\ X(\omega_6) = 60.$$

$\{20 \leqslant X < 40\}$ means that the outcomes appeared on the die are two or three.

For the experiment consists of measuring the lifetime of a person, let T_x denote the future lifetime of a person aged x and then $\{5 < T_x < 10)\}$ means the future lifetime of (x) is between five years and ten years.

The Definition and Properties of Distribution Function

That is to say, by means of random variable, we could use $\{X \in I\}$ denote random events where I is a subset of the real line, i.e.,

$$\{X \in I\} = \{\omega \mid X(\omega) \in I,\ \omega \in \Omega\},$$

which is different from the way we used in Chapter 1.
We can denote the probability that the values of X belong to the subset I by $P(X \in I)$. Then $P(X \in I)$ is equal to the probability that the outcome ω of the experiment will be such that $X(\omega) \in I$. In symbols,

$$P(X \in I) = P(\omega : X(\omega) \in I).$$

The Definition and Properties of Distribution Function

Example

Suppose that our experiment consists of tossing two fair coins. Let X denote the number of heads appearing. Then X is a random variable taking on one of the values 0, 1, 2 with respective probabilities

$$
\begin{aligned}
P(X = 0) &= P(\omega \mid X(\omega) = 0) = P(\{TT\}) = 1/4, \\
P(X = 1) &= P(\omega \mid X(\omega) = 1) = P(\{TH, HT\}) = 2/4, \\
P(X = 2) &= P(\omega \mid X(\omega) = 2) = P(\{HH\}) = 1/4.
\end{aligned}
$$

The Definition and Properties of Distribution Function

Example

It is obvious that,

$$P(\Omega) = P(X = 0) + P(X = 1) + P(X = 2) = 1.$$

We may also calculate the probabilities of $\{X \leqslant 1.5\}$, $\{-1 < X < 1\}$ and so on.

$$\begin{aligned} P(X \leqslant 1.5) &= P(\omega \mid X(\omega) \leqslant 1.5) \\ &= P(\{TT, TH, HT\}) \\ &= P(\{X = 0\} \cup \{X = 1\}) \\ &= P(X = 0) + P(X = 1) = 3/4. \end{aligned}$$

Similarly, $P(-1 < X < 1) = P(\{X = 0\}) = 1/4$. $\square$

The Definition and Properties of Distribution Function

The form of the subset I can be various. For any $x_1, x_2 \in \mathbb{R}$, I could be any of the following subsets:

$$(-\infty, x_1),\ (-\infty, x_1],\ \{x_1\},\ (x_1, x_2),\ (x_1, x_2],$$

$$[x_1, x_2],\ [x_1, x_2),\ [x_2, +\infty),\ (x_2, +\infty),\ \cdots.$$

Of course, we could know all the rules of probabilities of random variable if we could calculate the probability of any interval. But it seems impossible to calculate all above probabilities. In fact, to calculate the probabilities of $\{X \in I\}$, we just need to consider the following distribution function of random variable X.

The Definition and Properties of Distribution Function

Definition

The function $F(x)$ that associates with each real number x the probability $P(X \leqslant x)$ that the random variable X takes on a value smaller than or equal to this number is called the ***distribution function*** of X. *That is*

$$F(x) = P(X \leqslant x), \qquad \forall\, x \in \mathbb{R}. \tag{1}$$

The abbreviation for distribution function is d.f.. Some authors use the term ***cumulative distribution function***, instead of distribution function, and use the abbreviation c.d.f..

The Definition and Properties of Distribution Function

Property

The d.f. $F(x)$ of every random variable X has the following three properties.
(i) *It is nondecreasing as x increases; that is, if $x_1 < x_2$, then $F(x_1) \leqslant F(x_2)$.*
(ii) $\lim_{x \to -\infty} F(x) = 0$ *and* $\lim_{x \to +\infty} F(x) = 1$.
(iii) *It is always right-continuous; that is, $F(x) = F(x^+)$ at every point x.*

Proof. (i) If $x_1 < x_2$, then $\{X \leqslant x_1\} \subseteq \{X \leqslant x_2\}$. Hence, $P(X \leqslant x_1) \leqslant P(X \leqslant x_2)$.
(ii) These limiting values follow directly from the fact that $P(X \leqslant x)$ must approach 0 as $x \to -\infty$ and $P(X \leqslant x)$ must approach 1 as $x \to +\infty$. These relations can in turn be rigorously established by using the continuity of probability.

The Definition and Properties of Distribution Function

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(iii) *It is always right-continuous; that is, $F(x) = F(x^+)$ at every point x.*

Proof. (iii) Let $y_1 > y_2 > \cdots$ be a sequence of numbers that are decreasing such that $\lim_{n \to \infty} y_n = x$. Then the event $\{X \leqslant x\}$ is the intersection of all the events $\{X \leqslant y_n\}$ for $n = 1, 2, \cdots$. Hence, by the continuity of probability,

$$F(x) = P(X \leqslant x) = \lim_{n \to \infty} P(X \leqslant y_n) = F(x^+).$$

$\square$

Example

Are the following function d.f. of some random variables?

$$F_1(x) = \begin{cases} 0 & \text{for } x < -2, \\ 1/2 & \text{for } -2 \leqslant x < 0, \\ 2 & \text{for } x \geqslant 0, \end{cases}$$

Solution. Since $\lim_{x \to +\infty} F_1(x) = 2 > 1$, $F_1(x)$ can not be a d.f..

The Definition and Properties of Distribution Function

Example

Are the following function d.f. of some random variables?

$$F_2(x) = \begin{cases} 0 & \text{for } x < 0, \\ \sin x & \text{for } 0 \leqslant x < \pi, \\ 1 & \text{for } x \geqslant \pi, \end{cases}$$

Solution. Since $F_2(x) = \sin x$ is decreasing in the interval $(\pi/2, \pi)$ as x increases, $F_2(x)$ can not be a d.f..

Example

Are the following function d.f. of some random variables?

$$F_3(x) = \begin{cases} 0 & \text{for } x < 0, \\ x & \text{for } 0 \leqslant x < 1, \\ 1 & \text{for } x \geqslant 1. \end{cases}$$

Solution. The function $F_3(x)$ might be a d.f. of some random variable since it is satisfied the three conditions in Proposition. (In fact, it must be a d.f. of some random variable.)

Example

Suppose that the d.f. of a random variable X is

$$F(x) = \begin{cases} 0 & \text{for } x < -1, \\ a & \text{for } -1 \leqslant x < 1, \\ 2/3 - a & \text{for } 1 \leqslant x < 2, \\ a + b & \text{for } x \geqslant 2, \end{cases}$$

and $F(1) = 1/2$. Find the value of a and b.

Solution. Since $\lim_{x \to +\infty} F(x) = 1$, $a + b = 1$. Because the $d.f.$ must satisfy right-continuous property, $2/3 - a = 1/2$. So we have $a = 1/6, b = 5/6$. $\square$

The Definition and Properties of Distribution Function

Theorem

(i) *For every value* x, $P(X > x) = 1 - F(x)$.
(ii) *For all values* x_1 *and* x_2 *such that* $x_1 < x_2$,

$$P(x_1 < X \leqslant x_2) = F(x_2) - F(x_1).$$

(iii) *For each value* x, $P(X < x) = F(x^-)$.

Proof. (i) $P(X > x) = 1 - P(X \leqslant x) = 1 - F(x)$.
(ii)

$$\begin{aligned} P(x_1 < X \leqslant x_2) &= P(\{X \leqslant x_2\} - \{X \leqslant x_1\}) \\ &= P(X \leqslant x_2) - P(X \leqslant x_1) = F(x_2) - F(x_1). \end{aligned}$$

The Definition and Properties of Distribution Function

Theorem

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(iii) *For each value* x, $P(X < x) = F(x^-)$.

Proof. (iii) Let $y_1 < y_2 < \cdots$ be an increasing sequence of numbers such that $\lim_{n\to\infty} y_n = x$. Then

$$\{X < x\} = \bigcup_{n=1}^{\infty} \{X \leqslant y_n\}.$$

Thus

$$P(X < x) = \lim_{n\to\infty} P(X \leqslant y_n) = \lim_{n\to\infty} F(y_n) = F(x^-).$$

$\square$

The Definition and Properties of Distribution Function

Corralary

For every value x,

$$P(X = x) = F(x) - F(x^-) \qquad \text{and} \qquad P(X \geqslant x) = 1 - F(x^-).$$

For all values x_1 and x_2 such that $x_1 < x_2$,

$$P(x_1 < X < x_2) = F(x_2^-) - F(x_1),$$

$$P(x_1 \leqslant X < x_2) = F(x_2^-) - F(x_1^-),$$

and

$$P(x_1 \leqslant X \leqslant x_2) = F(x_2) - F(x_1^-).$$

The Definition and Properties of Distribution Function

Example

Suppose that the d.f. F of a random variable X is as sketched in the Fig. 2.4. Find each of the following probabilities:

(a) $P(X = -1)$	(b) $P(X < 0)$	(c) $P(X \leqslant 0)$
(d) $P(X = 1)$	(e) $P(0 < X \leqslant 3)$	(f) $P(0 < X < 3)$
(g) $P(0 \leqslant X \leqslant 3)$	(h) $P(1 < X \leqslant 2)$	(i) $P(1 \leqslant X \leqslant 2)$
(j)$P(X > 5)$	(k) $P(X \geqslant 5)$	(l) $P(3 \leqslant X \leqslant 4)$

The Definition and Properties of Distribution Function

Solution.

(a) $P(X = -1) = F(-1) - F(-1^-) = 0.1 - 0 = 0.1$,

(b) $P(X < 0) = F(0^-) = 0.1$,

(c) $P(X \leqslant 0) = F(0) = 0.2$,

(d) $P(X = 1) = F(1) - F(1^-) = 0.3 - 0.3 = 0$,

(e) $P(0 < X \leqslant 3) = F(3) - F(0) = 0.8 - 0.2 = 0.6$,

(f) $P(0 < X < 3) = F(3^-) - F(0) = 0.6 - 0.2 = 0.4$,

(g) $P(0 \leqslant X \leqslant 3) = F(3) - F(0^-) = 0.8 - 0.1 = 0.7$,

(h) $P(1 < X \leqslant 2) = F(2) - F(1) = 0.3 - 0.3 = 0$,

(i) $P(1 \leqslant X \leqslant 2) = F(2) - F(1^-) = 0.3 - 0.3 = 0$,

(j) $P(X > 5) = 1 - P(X \leqslant 5) = 1 - F(5) = 1 - 1 = 0$,

(k) $P(X \geqslant 5) = 1 - F(5^-) = 1 - 1 = 0$,

(l) $P(3 \leqslant X \leqslant 4) = F(4) - F(3^-) = 0.8 - 0.6 = 0.2$. $\square$

The Definition and Properties of Distribution Function

For one random variable X, the two things we must pay attention to are **the range** and **the rule** of X. The range of X means all the possible values that X could take. The rule of X means the d.f. of X. If the two things are known, then the probabilities that $\{X \in I\}$, for any $I \subseteq \mathbb{R}$ can be calculated.

Discrete Random Variable

For a discrete random variable X, we define the ***probability (mass) function*** $p(x)$ of X by $p(x) = P(X = x)$. The abbreviation for probability function is p.f..

x	x_1	x_2	$\cdots$	x_n	$(\cdots)$
$p(x)$	$p(x_1)$	$p(x_2)$	$\cdots$	$p(x_n)$	$(\cdots)$

Table: Tabular form of probability function

Let $\{x_1, x_2, \cdots\}$ be the set of possible values of the discrete random variable X. The function p has the following properties:

(i) $p(x_k) \geqslant 0$ for all x_k; $p(x) = 0$ for all other values of x,

(ii) $\sum_{k=1}^{\infty} p(x_k) = 1$.

Discrete Random Variable

Example

A sample space Ω containing n outcomes $\omega_1, \cdots, \omega_n$ is called a simple sample space if the probability assigned to each of the outcomes $\omega_1, \cdots, \omega_n$ is $1/n$. Let $X(\omega_i) = i/n$, $i = 1, 2, \cdots, n$. The p.f. of X is given by $p(i/n) = 1/n$, where $i = 1, 2, \cdots, n$. $\square$

x	$1/n$	$2/n$	$\cdots$	1
$1/n$	$1/n$	$1/n$	$1/n$	$1/n$

Discrete Random Variable

If X has a discrete distribution, the probability of each subset I of the real line can be determined from the relation

$$P(X \in I) = \sum_{x_k \in I} p(x_k). \tag{2}$$

The relationship between d.f. $F(x)$ and p.f. $p(x)$ of X is

$$F(x) = P(X \leqslant x) = \sum_{x_k \leqslant x} p(x_k) \tag{3}$$

Discrete Random Variable

Example

Suppose that the p.f. of random variable X is given by the following table. Find the values of the constant a in the table, $P(X > -1/2)$, and $P(-3 < X < 1)$.

x	-2	0	3	5
$p(x)$	$1/4$	a	$1/2$	$1/12$

Solution. Since $1/4 + a + 1/2 + 1/12 = 1$, $a = 1/6$.
$P(X > -1/2) = P(X = 0) + P(X = 1) + P(X = 3) = 1/6 + 1/2 + 1/12 = 3/4$.
$P(-3 < X < 1) = P(X = -2) + P(X = 0) = 1/4 + 1/6 = 5/12$. $\square$

Discrete Random Variable

Example

Consider the corresponding p.f. is given by the table

x	0	1	2
$p(x)$	1/4	2/4	1/4

Find the d.f. of the random variable X.

Solution. If $x < 0$, then $F(x) = P(X \leqslant x) = P(\emptyset) = 0$.
If $0 \leqslant x < 1$, then $F(x) = P(X \leqslant x) = P(X = 0) = 1/4$.
If $1 \leqslant x < 2$, then

$$F(x) = P(X \leqslant x) = P(X = 0 \text{ or } 1)$$

$$= P(X = 0) + P(X = 1) = 1/4 + 2/4 = 3/4.$$

Discrete Random Variable

Solution. If $x \geqslant 2$, then

$$F(x) = P(X \leqslant x) = P(X = 0,\ 1 \text{ or } 2)$$

$$= P(X = 0) + P(X = 1) + P(X = 2) = 1.$$

That is,

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ 1/4 & \text{for } 0 \leqslant x < 1, \\ 3/4 & \text{for } 1 \leqslant x < 2, \\ 1 & \text{for } x \geqslant 2. \end{cases}$$

$\square$

Discrete Random Variable

Example

If the d.f. of random variable X is the the same as in above example. Find the p.f. of random variable X.

Solution. When $x \in (-\infty, 0)$, $(0, 1)$, $(1, 2)$ or $(2, +\infty)$, respectively, we have

$$P(X = x) = F(x) - F(x^-) = 0.$$

When $x = 0$, $P(X = 0) = F(0) - F(0^-) = 1/4 - 0 = 1/4$.
When $x = 1$, $P(X = 1) = F(1) - F(1^-) = 3/4 - 1/4 = 1/2$.
When $x = 2$, $P(X = 2) = F(2) - F(2^-) = 1 - 3/4 = 1/4$. $\square$

For discrete random variable, using p.f. is more straightforward than using d.f. in many cases. Because we can easily know all possible values taken by X and the corresponding probabilities.

Continuous Random Variable

Definition

We call X a ***continuous random variable*** *if there is a function f defined for all $x \in \mathbb{R}$ and having the following properties:*
(i) $f(x) \geqslant 0$ *for any real number x,*
(ii) *if I is any subset of $\mathbb{R}$, then*

$$P(X \in I) = \int_I f(x)\, dx,$$

where $f(x)$ is called the ***probability density function*** *of r.v. X. (The abbreviation for probability density function is p.d.f.)*

Continuous Random Variable

If f is the p.d.f. of the r.v. X, then

$$P(a \leqslant X \leqslant b) = \int_a^b f(x)dx.$$

The probability is the area, and the area is the integral from a to b.
If $I = (-\infty, x]$, then the distribution function F of X is given by

$$F(x) = P(X \leqslant x) = \int_{-\infty}^{x} f(x)\ dx. \tag{4}$$

We also deduce that

$$\frac{d}{dx}F(x) = f(x) \tag{5}$$

for any x where $F(x)$ is differentiable.

Continuous Random Variable

Remark

(i) *The d.f.* $F(x)$ *of continuous r.v.* X *is continuous since* $P(X = x) = 0$.
(ii) *By using equation (4) and* $F(\infty) = 1$, *we get*

$$\int_{-\infty}^{\infty} f(x)\, dx = F(\infty) = 1. \tag{6}$$

To be a p.d.f. of some continuous random variable, the nonnegative function $f(x)$ *need satisfy equation (6).*

Continuous Random Variable

Remark

(iii) *The density function is different from the probability function p of a discrete random variable. Indeed, f does not give the probability that the random variable X takes on the value x. Moreover, we may have that $f(x) > 1$. Actually, we may write that*

$$\begin{aligned} f(x) = \frac{d}{dx}F(x) &= \lim_{\triangle x \to 0} \frac{F(x+\triangle x) - F(x)}{\triangle x} \\ &= \lim_{\triangle x \to 0} \frac{P(x < X \leqslant x + \triangle x)}{\triangle x}. \end{aligned}$$

Thus, $f(x)\triangle x$ is approximately equal to the probability that X takes on a value in an interval of length $\triangle x$ about x. A probability density acts like an ordinary density (like grams per centimeter). Even though there is no mass exactly at a particular point, the total mass can be found by integrating.

Continuous Random Variable

Example

Consider the function $f(x) = 2 - 2x$ for $0 \leqslant x \leqslant 1$.
(a) Prove $f(x)$ is indeed a p.d.f..
(b) Find the corresponding d.f. $F(x)$.
(c) Calculate the value of $P(0.3 \leqslant X \leqslant 0.6)$.

Solution. (a) First, we check that this is indeed a p.d.f. because

$$\int_0^1 f(x)dx = \int_0^1 (2 - 2x)dx = (2x - x^2)\Big|_0^1 = 1.$$

Although the nonnegative function itself takes on values greater than 1, the total area is equal to 1.

Continuous Random Variable

Solution. (b) The d.f. is if $x < 0$, then $F(x) = 0$.
If $x > 1$, then $F(x) = 1$.
If $0 \leqslant x \leqslant 1$, then
$F(x) = \int_0^x f(y)dy = \int_0^x (2-2y)dy = (2y - y^2)\Big|_0^x = 2x - x^2.$
(c) With the p.d.f. and d.f. above, we can find the probability that x is between 0.3 and 0.6 either by integrating the p.d.f. or by evaluating the d.f. at the endpoints and subtracting. In the first case, we find

$$P(0.3 \leqslant X \leqslant 0.6) = \int_{0.3}^{0.6} (2-2x)dx = (2x - x^2)\Big|_{0.3}^{0.6} = 0.33.$$

Alternatively, $P(0.3 \leqslant X \leqslant 0.6) = F(0.6) - F(0.3) = 0.33.$ $\square$

Continuous Random Variable

Example

Suppose that the d.f. of random variable X is

$$F(x) = \begin{cases} A + Be^{-x^2/2} & \text{for } x > 0, \\ 0 & \text{for } x \leqslant 0. \end{cases}$$

Calculate (a) the values of A and B; (b) $f(x)$; (c) $P(1 < X < 2)$.

Solution. (a) Since $F(+\infty) = 1$, $A = 1$. Since $F(0) = F(0^+)$, $A + B = 0$. Thus we can obtain that $B = -1$.

Continuous Random Variable

Solution. (a) Since $F(+\infty) = 1$, $A = 1$. Since $F(0) = F(0^+)$, $A + B = 0$. Thus we can obtain that $B = -1$.
(b) By using equation (5), we have

$$f(x) = F^{'}(x) = \begin{cases} xe^{-x^2/2} & \text{for } x > 0, \\ 0 & \text{for } x \leqslant 0. \end{cases}$$

(c) $P(1 < X < 2) = F(2) - F(1) = 0.47$.
Alternatively, $P(1 < X < 2) = \int_1^2 xe^{-x^2/2} dx = 0.47$. $\square$

Continuous Random Variable

Example

The exact p.d.f. for the time T when a diffusing toxic molecule leaves the cell is

$$f(t) = \begin{cases} c\ e^{-0.1t} & \text{for } t > 0, \\ 0 & \text{for } t \leqslant 0. \end{cases}$$

when the rate of departure is 0.1. Find
(a) the value of c,
(b) the probability that the Molecule left between 0 and time 10,
(c) the probability that the Molecule left between 2.0 and time 2.1.

Continuous Random Variable

Solution. (a) Since $\int_{-\infty}^{+\infty} f(t)dt = 1$,

$$\int_{-\infty}^{+\infty} f(t)dt = \int_{-\infty}^{0} 0dt + \int_{0}^{+\infty} c\, e^{-0.1t}dt = 1 \Rightarrow \frac{c}{0.1} = 1 \Rightarrow c = 0.1.$$

(b) $P(0 \leqslant T \leqslant 10) = \int_{0}^{10} 0.1e^{-0.1t}dt = -e^{-0.1t}\Big|_{0}^{10} =$
$-e^{-1.0} + e^{0} = 0.632.$

If we ran this experiment 1000 times, then about 632 molecules would leave before time 10.

Continuous Random Variable

Solution. (c) $P(2.0 \leqslant T \leqslant 2.1) = \int_{2.0}^{2.1} 0.1e^{-0.1t}dt =$
$-e^{-0.1t}\Big|_{2.0}^{2.1} = -e^{-0.21} + e^{-0.2} = 0.00815.$

Because the interval is narrow, we can approximate this region as a rectangle with base 0.1 and height $f(2.0)$.

$$P(2.0 \leqslant T \leqslant 2.1) \approx base \times height = 0.1 \times f(2.0)$$

$$= 0.2 \cdot 0.1e^{-0.2} = 0.00819.$$

The probability that it leaves near time 2.0 is proportional to the height of probability of the p.d.f. at that point. $\square$

Section 2.2 The Distribution Function of a Random Variable

- 2.2.1 The Definition and Properties of Distribution Function
- 2.2.2 The Distribution Function of Function of a Random Variable

The Distribution Function of Function of a Random Variable

Because a random variable is a real-valued function and the composition of two functions is also a function, we can assert that if X is a random variable, then $Y := g(X) = g(X(\omega))$, where g is a real-valued function defined on the real line, is a random variable as well. In this section, we show how to obtain the probability function or the density function of Y.

Let us see discrete case first. Here we mainly discuss the case X and $Y = g(X)$ are both discrete.

The Distribution Function of Function of a Random Variable

Example

Let X be a discrete random variable whose probability function is given in Table

x	-1	0	1
$p_X(x)$	$1/4$	$1/4$	$1/2$

Let $Y = 2X$ and $W = X^2$. Find the probability functions of Y and W respectively.

The Distribution Function of Function of a Random Variable

Solution. Because the function $g : x \to 2x$ is bijective (i.e., to a given $y = g(x) = 2x$, there corresponds one and only one x and vice versa), the number of possible values of the random variable Y will be the same as the number of possible values of X. We find that

$$p_Y(-2) = P(Y = -2) = P(X = -1) = 1/4,$$
$$p_Y(0) = P(Y = 0) = P(X = 0) = 1/4,$$
$$p_Y(2) = P(Y = 2) = P(X = 1) = 1/2.$$

We can describe the p.f. of Y by

x	-1	0	1
y	-2	0	2
$p_Y(y)$	$1/4$	$1/4$	$1/2$

The Distribution Function of Function of a Random Variable

Solution. For $W = X^2$, the possible values of the random variable of W is 0 and 1. And

$$p_W(0) = P(W = 0) = P(X = 0) = 1/4,$$

$$p_W(1) = P(W = 1) = P(\{X = -1\} \cup \{X = 1\})$$
$$= P(X = -1) + P(X = 1) = 3/4.$$

The p.f. of Y can be described by Table

w	1	0	1
x	-1	0	1
$p_Y(y)$	1/4	1/4	1/2

$\Longrightarrow$

w	0	1
$p_W(x)$	1/4	3/4

The Distribution Function of Function of a Random Variable

In the case when the random variable X can take a (countably) infinite number of values, we apply the transformation to an arbitrary value of X, and we can find a general formula for the function $p_Y(y)$. That is,

$$p_Y(y) = P(Y = y) = P(g(X) = y) = \sum_{x:g(x)=y} p(x). \tag{7}$$

The Distribution Function of Function of a Random Variable

Example

Suppose that the p.f. of X is

$$p_X(x) = (1-p)^{x-1}p, \qquad x = 1, 2, \cdots$$

where $0 < p < 1$. Let $Y = X^2$. Because the value of X is $1, 2, \cdots$, the quadratic function is (here) a bijective transformation and we easily calculate

$$p_Y(y) = P(Y = y) = P(X^2 = y)$$

$$= p_X(\sqrt{y}) = (1-p)^{\sqrt{y}-1}p \quad \text{for } y = 1, 4, 9, \cdots.$$

$\square$

The Distribution Function of Function of a Random Variable

Next, let us consider continuous case. Here we assume that X and $Y = g(X)$ are both continuous random variables.

Example

Let X be a continuous nonnegative random variable with density function

$$f_X(x) = \begin{cases} \dfrac{x}{8} & \text{for } 0 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

and let $Y = 2X + 1$. Find $f_Y(y)$, the probability density function of Y.

The Distribution Function of Function of a Random Variable

Solution. First we find the d.f. of the random variable Y.

$$\begin{aligned} F_Y(y) =& P(Y \leqslant y) = P(2X+1 \leqslant y) \\ =& P(X \leqslant \frac{y-1}{2}) = \int_{-\infty}^{\frac{y-1}{2}} f_X(x)dx. \end{aligned}$$

If $y \leqslant 1$, $\dfrac{y-1}{2} \leqslant 0$, then $F_Y(y) = 0$.

If $1 < y < 9$, $0 \leqslant \dfrac{y-1}{2} \leqslant 4$, then

$$F_Y(y) = \int_0^{\frac{y-1}{2}} \frac{x}{8} dx = \frac{(y-1)^2}{64}.$$

The Distribution Function of Function of a Random Variable

Solution. If $y \geqslant 9$, then $F_Y(y) = 1$. That is,

$$F_Y(y) = \begin{cases} 0 & \text{for } y \leqslant 1, \\ \dfrac{(y-1)^2}{64} & \text{for } 1 < y < 9, \\ 1 & \text{for } y \geqslant 9. \end{cases}$$

Hence, by differentiating, we obtain

$$f_Y(y) = \begin{cases} \dfrac{y-1}{32} & \text{for } 1 < y < 9, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

The Distribution Function of Function of a Random Variable

Example

Suppose that X is a continuous random variable with probability density $f_X(x)$. Find the probability density function $f_Y(y)$ of $Y = X^2$.

Solution. For $y \geqslant 0$,

$$F_Y(y) = P(Y \leqslant y) = P(X^2 \leqslant y)$$

$$= P(-\sqrt{y} \leqslant X \leqslant \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

Differentiation yields

$$f_Y(y) = \frac{1}{2\sqrt{y}}\Big[f_X(\sqrt{y}) + f_X(-\sqrt{y})\Big].$$

$\square$

The Distribution Function of Function of a Random Variable

Example

If X has a probability density function f_X, then $Y = |X|$ has a density function that is obtained as follows. For $y \geqslant 0$, we have

$$F_Y(y) = P(Y \leqslant y) = P(|X| \leqslant y)$$

$$= P(-y \leqslant X \leqslant y) = F_X(y) - F_X(-y).$$

Hence, by differentiating, we obtain

$$f_Y(y) = f_X(y) + f_X(-y), \qquad y \geqslant 0.$$

$\square$

The Distribution Function of Function of a Random Variable

The method employed above, **the distribution function method**, is the usual way to obtain p.d.f. of a function of a continuous random variable. This method can be divided into four steps:
(i) transform the event $\{Y \leqslant y\}$ to $\{g(X) \leqslant y\}$,
(ii) transform the event $\{g(X) \leqslant y\}$ to $\{X \in I_y\}$, where $I_y = \{x : g(x) \leqslant y\}$,
(iii) calculate the probability $P(X \in I_y)$, which is just the d.f. $F(y)(= P(Y \leqslant y))$ of Y,
(iv) obtain p.d.f. $f(y)$ of Y by $f(x) = F^{'}(x)$.
The distribution function method can also be used to prove the following theorem.

The Distribution Function of Function of a Random Variable

Theorem

Let X be a continuous random variable having p.d.f. f_X. Suppose that $g(x)$ is a strictly monotonic (increasing or decreasing), differentiable (and thus continuous) function of x. Then the random variable Y defined by $Y = g(X)$ has a p.d.f. given by

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y))\left|\dfrac{d}{dy}g^{-1}(y)\right| & \text{if } y = g(x) \text{ for some } x, \\ 0 & \text{if } y \neq g(x) \text{ for all } x, \end{cases} \tag{8}$$

where $x = g^{-1}(y)$ is defined to equal that value of x such that $g(x) = y$.

The Distribution Function of Function of a Random Variable

Proof. Suppose that $g(x)$ is an increasing function. Suppose that $y = g(x)$ for some x. Then, with $Y = g(X)$,

$$F_Y(y) = P(g(X) \leqslant y) = P(X \leqslant g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiation gives

$$f_Y(y) = f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y),$$

since $g^{-1}(y)$ is nondecreasing, so its derivative is nonnegative.

When $y \neq g(x)$ for any x, then $F_Y(y)$ is either 0 or 1, and in either case $f_Y(y) = 0$. $\square$

The Distribution Function of Function of a Random Variable

Example

Let X be a continuous nonnegative random variable with density function

$$f_X(x) = \begin{cases} \dfrac{x}{8} & \text{for } 0 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

and let $Y = 2X + 1$. Find $f_Y(y)$ by using Theorem.

Solution. If $y = g(x) = 2x + 1$, then

$$g^{-1}(y) = \frac{y-1}{2} \quad \text{and} \quad \frac{d}{dy} g^{-1}(y) = \frac{1}{2}.$$

Hence,

$$f_Y(y) = \begin{cases} \frac{1}{2} \cdot \frac{\frac{y-1}{2}}{8} & \text{for } 0 < \frac{y-1}{2} < 4 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} \frac{y-1}{32} & \text{for } 1 < y < 9, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

The Distribution Function of Function of a Random Variable

Example

Let X be a continuous nonnegative random variable with density function f_X, and let $Y = X^n$. Find f_Y, the probability density function of Y.

Solution. If $g(x) = x^n$, then

$$g^{-1}(y) = y^{1/n}$$

and

$$\frac{d}{dy} g^{-1}(y) = \frac{1}{n} y^{1/n-1}.$$

Hence,

$$f_Y(y) = \frac{1}{n} y^{1/n-1} f_X(y^{1/n}).$$

For $n = 2$, this gives $f_Y(y) = \frac{1}{2\sqrt{y}} f_X(\sqrt{y})$. $\square$

The Distribution Function of Function of a Random Variable

In the end of this section, we introduce another method, **the density function method**, which finds f_Y directly, and then F_Y by integration. Consider above Example ($n = 2$) again.

First, we have $f_Y(y)|dy| = f_X(\sqrt{y})dx + f_X(-\sqrt{y})dx$. We write $|dy|$ because probabilities are never negative. Then

$$f_Y(y) = \frac{f_X(\sqrt{y})}{|dy/dx|_{x=\sqrt{y}}} + \frac{f_X(-\sqrt{y})}{|dy/dx|_{x=-\sqrt{y}}}.$$

With $y = x^2$, $dy/dx = 2x$, so

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}\Big[f_X(\sqrt{y}) + f_X(-\sqrt{y})\Big].$$

Section 2.3 Mathematical Expectation and Variance

- 2.3.1 Expectation of a Random Variable
- 2.3.2 Expectation of Functions of a Random Variable
- 2.3.3 Variance of a Random Variable
- 2.3.4 The Application of Expectation and Variation

Expectation of a Random Variable

The distribution of a random variable X contains all of the probabilistic information about X.

But many random variables have complicated distribution functions and it is therefore difficult to obtain an intuitive understanding of the behavior of the random variable by simply knowing the distribution function.

Expectation of a Random Variable

Summaries of the distribution, such as the average value, or expected value, can be useful for giving people an idea of where we expect X to be without trying to describe the entire distribution.

For example, when a large collection of numbers is assembled, as in a census, we are usually interested not in the individual numbers, but rather in certain descriptive quantity such as the average. In this section, we mainly introduce two measures: the expectation and the variance.

Expectation of a Random Variable

Definition

The ***expectation*** *(the* ***mean*** *or the* ***expected value****) of a random variable X is given by*

$$\mu := E(X) = \begin{cases} \displaystyle\sum_{i=1}^{\infty} x_i p(x_i) & \textit{if } X \textit{ is discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x f(x) dx & \textit{if } X \textit{ is continuous.} \end{cases} \tag{9}$$

In words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it.

Expectation of a Random Variable

For example, let us consider the random experiment of rolling a die. If an odd number turns up, then we win an amount equal to this number; if an even number turns up, then we lose an amount equal to this number. For example, if a two turns up we lose 2, and if a three comes up we win 3. We want to decide if this is a reasonable game to play. We first try simulation. The program Die carries out this simulation. The program prints the frequency and the relative frequency with which each outcome occurs. It also calculates the average winnings. We have run the program twice. The results are shown in Table

Expectation of a Random Variable

Winning	$n = 100$		$n = 10000$	
	Frequency	Relative Frequency	Frequency	Relative Frequency
1	17	0.17	1681	0.1681
-2	17	0.17	1678	0.1678
3	16	0.16	1626	0.1626
-4	18	0.18	1696	0.1696
5	16	0.16	1686	0.1686
-6	16	0.16	1633	0.1633

Table: Frequencies for dice game

Expectation of a Random Variable

In the first run we have played the game 100 times. In this run our average gain is -0.57. It looks as if the game is unfavorable, and we wonder how unfavorable it really is. To get a better idea, we have played the game 10,000 times. In this case our average gain is -0.4949.
We note that the relative frequency of each of the six possible outcomes is quite close to the probability $1/6$ for this outcome. This corresponds to our frequency interpretation of probability. It also suggests that for very large numbers of plays, our average gain should be

$$1 \cdot \frac{1}{6} - 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} - 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} - 6 \cdot \frac{1}{6} = -0.5.$$

This agrees quite well with our average gain for 10,000 plays.

Expectation of a Random Variable

We note that the value we have chosen for the average gain is obtained by taking the possible outcomes, multiplying by the probability, and adding the results. This suggests the above definition for the expected outcome of an experiment.

However, the idea of adding the value times the probability to find the expectation does not quite work for continuous random variables because each simple event has probability 0. By returning to the Riemann sum approach, we will find the continuous case of equation (9) in Definition is reasonable.

Expectation of a Random Variable

By breaking the set of all possible values of the random variable into pieces, we can approximate a continuous random variable with discrete random variable. We approximate the values of the continuous random variable with a discrete random variable that takes on the values $x_0, x_1, \cdots x_{n-1}$ with probabilities $f(x_0)\triangle x, f(x_1)\triangle x, \cdots, f(x_{n-1})\triangle x$. Scientifically, this corresponds to being unable to distinguish the values in the interval x_i to $x_i + \triangle x$ from x_i itself. The expectation of the approximate discrete random variable is

$$\sum_{i=0}^{n-1} x_i f(x_i)\triangle x.$$

This has precisely the form of a Riemann sum. Taking the limit as $n \to \infty$ and $\triangle x \to 0$ gives a definite integral, leading us to the definition of the expectation of a continuous random variable.

Expectation of a Random Variable

Example

Find $E(X)$, where the p.f. of X is $p(0) = 0.1$ and $p(1) = 0.9$.

Solution. Since $p(0) = 0.1$ and $p(1) = 0.9$, we obtain

$$E(X) = 0 \cdot 0.1 + 1 \cdot 0.9 = 0.9.$$

$\square$

Remark

The probability concept of expectation is analogous to the physical concept of the center of gravity of a distribution of a mass. Consider a discrete random variable X having probability function $p(x_i), i \geqslant 1$. If we now imagine a weightless rod in which weights with mass $p(x_i), i \geqslant 1$, are located at the points $x_i, i \geqslant 1$, then the point at which the rod would be in balance is known as the center of gravity.

Expectation of a Random Variable

Example

Suppose that you plan on starting a new business and would like to know the expected return for the business. You develop a subjective probability distribution of the returns and their associated probabilities. Note: X=return in \$1,000/year (e.g., -10 means a loss of \$10,000).

x	-10	0	10	20
$p(x)$	0.20	0.25	0.40	0.15

What is the expected return for this business?

Expectation of a Random Variable

Solution.

$$\begin{aligned} E(X) &= \sum x \cdot p(x) \\ &= x_1 P(x_1) + x_2 P(x_2) + x_3 P(x_3) + x_4 P(x_4) \\ &= -10 \cdot (0.2) + 0 \cdot (0.25) + 10 \cdot (0.4) + 20 \cdot (0.15) \\ &= -2 + 0 + 4 + 3 = 5. \end{aligned}$$

The expected return for this business is \$5,000. $\square$

Expectation of a Random Variable

Example

Mr. Smith would like to buy catastrophic insurance against a heart attack. The insurance will pay Mr. Smith \$100,000 if he has a heart attack. The probability that Mr. Smith will have a heart attack in a given year is 0.01, so the probability that he will not have a heart attack is 0.99. The insurance premium for the heart attack policy is \$800 per year.

(a) What is the expected return for the insurance company?

(b) A company officer asks the premium manager to determine the annual premium required for the insurance company to make \$150 on each \$100,000 policy. Perform the calculations to determine the annual premium.

Expectation of a Random Variable

Solution.
(a)

$$E(X) = \sum x \cdot p(x)$$

$$800 \cdot 0.99 + (-100000 + 800) \cdot 0.01 = -200.$$

The expected return for the insurance company is negative. That means the insurance company is losing money on heart attack insurance.
(b) Let k be the required annual premium. Since

$$E(X) = \sum x \cdot p(x),$$

$$150 = k \cdot 0.99 + (-100000 + k) \cdot 0.01.$$

So the annual premium k is \$1150. $\square$

Expectation of a Random Variable

Example

A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the students is randomly chosen. Let X denote the number of students on the bus of that randomly chosen student, and find $E(X)$.

Solution. Since the randomly chosen student is equally likely to be any of the 120 students, it follows that

$$P(X = 36) = \frac{36}{120}, \quad P(X = 40) = \frac{40}{120}, \quad P(X = 44) = \frac{44}{120}.$$

Hence,

$$E(X) = 36 \cdot \frac{3}{10} + 40 \cdot \frac{1}{3} + 44 \cdot \frac{11}{30} = \frac{1208}{30} = 40.2667.$$

Expectation of a Random Variable

Solution. However, the average number of students on a bus is $120/3 = 40$, showing that the expected number of students on the bus of a randomly chosen student is larger than the average number of students there are on a bus, the more likely it is that randomly chosen student would have been on that bus. As a result, buses with many students are given more weight than those with fewer students. $\square$

Expectation of a Random Variable

Example

Suppose that we toss a fair coin until a head first comes up, and let X represent the number of tosses which were made. Calculate the value of $E(X)$.

Solution. The possible values of X are $1, 2, \cdots$, and the distribution function of X is

$$p(i) = \frac{1}{2^i}, \quad i = 1, 2, \cdots .$$

Thus, we have

$$\begin{aligned} E(X) &= \sum_{i=1}^{\infty} i \frac{1}{2^i} = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=2}^{\infty} \frac{1}{2^i} + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots = 2. \end{aligned}$$

$\square$

Expectation of a Random Variable

If X can have a finite number of different values, then there will be only a finite number of terms in the summation. However, if there is an infinite sequence of different possible values of X, then the summation in equation (9) consists of an infinite series of terms. Such a series may not converge for a given p.f.. It is said that the expectation $E(X)$ exists if and only if the summation in equation (9) is absolutely convergent, that is, if and only if

$$\sum_{all\ \ x} |x| f(x) < \infty \quad \text{or} \quad \int_x |x| f(x) dx < \infty. \tag{10}$$

In other words, if relation (10) is satisfied, then $E(X)$ exists and its value is given by equation (9). If relation (10) is not satisfied, then $E(X)$ does not exists.

Expectation of a Random Variable

Example

Suppose that we flip a coin until a head first appears, and if the number of tosses equals n, then we are paid 2^n dollars. What is the expected value of the payment?

Solution. We let Y represent the payment. Then,

$$P(Y = 2^n) = \frac{1}{2^n}$$

for $n \geqslant 1$. Thus,

$$E(Y) = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n}$$

which is a divergent sum. Thus, Y has no expectation. $\square$

Expectation of a Random Variable

This example is called the ***St. Petersburg Paradox***. The fact that the above sum is infinite suggests that a player should be willing to pay any fixed amount per game for the privilege of playing this game. The reader is asked to consider how much he or she would be willing to pay for this privilege. It is unlikely that the reader's answer is more than 10 dollars; therein lies the paradox.

Expectation of a Random Variable

In the early history of probability, various mathematicians gave ways to resolve this paradox. One idea (due to G. Cramer) consists of assuming that the amount of money in the world is finite. He thus assumes that there is some fixed value of n such that if the number of tosses equals or exceeds n, the payment is 2^n dollars. The reader is asked to show in Exercise 2.24 that the expected value of the payment is now finite.

Expectation of a Random Variable

Daniel Bernoulli and Cramer also considered another way to assign value to the payment. Their idea was that the value of a payment is some function of the payment; such a function is now called a utility function. Examples of reasonable utility functions might include the square-root function or the logarithm function. In both cases, the value of 2^n dollars is less than twice the value of n dollars. It can easily be shown that in both cases, the expected utility of the payment is finite (see Exercise 2.24).

Expectation of a Random Variable

Now let us see some examples of continuous case.

Example

Suppose that the p.d.f. of a random variable X is

$$f(x) = \begin{cases} cx & \text{for } 0 < x < 4, \\ 0 & \text{otherwise,} \end{cases}$$

where c is a given constant. Determine the value of c and the expected value of X.

Expectation of a Random Variable

Solution. For every p.d.f., it must be true that $\int_{-\infty}^{+\infty} f(x)dx = 1$. Therefore, in this example,

$$\int_0^4 cxdx = 8c = 1.$$

Hence, $c = 1/8$. It follows that

$$E(X) = \int_0^4 x \cdot \frac{1}{8} xdx = \frac{8}{3}.$$

$\square$

Expectation of a Random Variable

Example

Suppose that the d.f. of a random variable X is as follows:

$$F(x) = \begin{cases} e^{x-3} & \text{for } x \leqslant 3, \\ 1 & \text{for } x > 3. \end{cases}$$

Determine the expected value of X.

Solution. Of course we can find the p.d.f. of X from d.f., then $E(X)$ can be obtained by using the formula in Definition. But another way is using d.f. directly to get $E(X)$.

$$E(X) = \int_{-\infty}^{+\infty} x dF(x) = \int_{-\infty}^{3} x de^{x-3}$$

$$= xe^{x-3}\Big|_{-\infty}^{3} - e^{x-3}\Big|_{-\infty}^{3} = 3 - 1 = 2.$$

$\square$

Expectation of a Random Variable

Example

Suppose that X is a continuous random variable and the p.d.f of X is

$$f(x) = \frac{1}{\pi}\frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

This distribution is called the ***Cauchy distribution***. $E(X)$ does not exist since

$$\int_{-\infty}^{\infty} |x| f(x) dx = 2\int_0^{\infty} \frac{1}{\pi}\frac{x}{1+x^2} dx = \frac{1}{\pi}\ln(1+x^2)\Big|_0^{\infty} \text{ diverges.}$$

$\square$

Expectation of a Random Variable

Property

(i) $E(c) = c$, *for any constant* c.
(ii) *If there exists a constant such that* $P(X \geqslant a) = 1$, *then* $E(X) \geqslant a$. *If there exists a constant such that* $P(X \leqslant b) = 1$, *then* $E(X) \leqslant b$.

Proof. The proof of (i) is obvious.
(ii) For convenience, we assume that X has a continuous distribution for which the p.d.f. is f, and we shall suppose first that $P(X \geqslant a) = 1$. Then

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{a}^{\infty} x f(x) dx \\ &\geqslant \int_{a}^{\infty} a f(x) dx = aP(X \geqslant a) = a. \end{aligned}$$

$\square$

Expectation of a Random Variable

From Proposition, we can get

Corralary

(i) *If* $P(a \leqslant X \leqslant b) = 1$, *then* $a \leqslant E(X) \leqslant b$.

(ii) *If* $P(X \geqslant a) = 1$ *and* $E(X) = a$, *then it must be true that* $P(X > a) = 0$ *and* $P(X = a) = 1$.

Section 2.3 Mathematical Expectation and Variance

- 2.3.1 Expectation of a Random Variable
- 2.3.2 Expectation of Functions of a Random Variable
- 2.3.3 Variance of a Random Variable
- 2.3.4 The Application of Expectation and Variation

Expectation of Functions of a Random Variable

Suppose that we are given a random variable along with its probability (density) function and that we want to compute the expected value of some function of X, say, $g(X)$.

How can we accomplish this?

One way is as follows: suppose that X and $g(X)$ are both discrete (or continuous), then $g(X)$ has a probability (density) function, which can be determined from the probability (density) function of X. Once we have determined the probability (density) function of $g(X)$, we can compute $E[g(X)]$ by using the definition of expected value.

Expectation of Functions of a Random Variable

Example

Let X denote a random variable that takes on any of the values $-1, 0$, and 1 with respective probabilities

$$P(X = -1) = 0.2 \quad P(X = 0) = 0.5 \quad P(X = 1) = 0.3$$

Compute $E(X^2)$.

Solution. Let $Y = X^2$. Then the probability mass function of Y is given by

$$\begin{aligned} P(Y = 1) &= P(X = -1) + P(X = 1) = 0.5 \\ P(Y = 0) &= P(X = 0) = 0.5. \end{aligned}$$

Hence, $E(Y) = E(X^2) = 1 \cdot 0.5 + 0 \cdot 0.5 = 0.5.$ $\square$

Expectation of Functions of a Random Variable

Example

Let X be a continuous nonnegative random variable with density function

$$f_X(x) = \begin{cases} \dfrac{x}{8} & \text{for } 0 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

and let $Y = 2X + 1$. Compute $E(Y)$.

Solution. We have known the p.d.f. of Y. So

$$E(Y) = \int_{-\infty}^{+\infty} y f_Y(y) dy = \int_1^9 y \cdot \frac{y-1}{32} dy = \frac{19}{3}.$$

$\square$

Expectation of Functions of a Random Variable

As we know, sometimes it is difficult to determine the p.f. or p.d.f. of $g(X)$. In fact, we can compute $E[g(X)]$ by the the p.f. (or p.d.f.) of X directly.

Property

The mathematical expectation of a function $g(X)$ of a random variable X can be calculated by

$$E[g(X)] = \begin{cases} \sum_{i=1}^{\infty} g(x_i)p(x_i) & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} g(x)f(x)dx & \text{if } X \text{ is continuous.} \end{cases} \tag{11}$$

The expectation $E[g(X)]$ will exist if and only if

$$\int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty.$$

Expectation of Functions of a Random Variable

Example

Let X be a continuous nonnegative random variable with density function

$$f_X(x) = \begin{cases} \dfrac{x}{8} & \text{for } 0 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

and let $Y = 2X + 1$. Compute $E(Y)$.

Solution.

$$E(Y) = E(2X+1) = \int_{-\infty}^{+\infty} (2x+1) f(x) dx = \int_0^4 (2x+1) \cdot \frac{x}{8} dx = \frac{19}{3}.$$

$\square$

Expectation of Functions of a Random Variable

Proof of Proposition As in the preceding verification, we will group together all the terms in $\sum_i g(x_i)p(x_i)$ having the same value of $g(x_i)$. Specifically, suppose that $y_j, j \geqslant 1$, represent the different values of $g(x_i), i \geqslant 1$. Then, grouping all the $g(x_i)$ having the same value gives

$$\begin{aligned}
\sum_i g(x_i)p(x_i) &= \sum_j \sum_{i:g(x_i)=y_j} g(x_i)p(x_i) \\
&= \sum_j \sum_{i:g(x_i)=y_j} y_j p(x_i) \\
&= \sum_j y_j \sum_{i:g(x_i)=y_j} p(x_i) \\
&= \sum_j y_j P(g(X) = y_j) \\
&= E(g(X)).
\end{aligned}$$

$\square$

Expectation of Functions of a Random Variable

Example

Suppose that the p.d.f. of wind velocity V is

$$f(v) = \begin{cases} \dfrac{1}{a} & \text{for } 0 < v < a, \\ 0 & \text{otherwise,} \end{cases}$$

and the pressure on wing W is the function of wind velocity V, that is, $W = g(V) = kV^2$, where k is a constant. Determine the expected value of W.

Solution.

$$E(W) = \int_{-\infty}^{+\infty} kv^2 \cdot f(v)dv = \int_0^a kv^2 \cdot \frac{1}{a} dv = \frac{1}{3}ka^2.$$

$\square$

Expectation of Functions of a Random Variable

Property

Suppose that $g(X)$ is a function of the random variable X. Then for all constants a and b,

$$E[ag(X) + b] = aE[g(X)] + b.$$

Specially, if $g(X) = X$, then

$$E(aX + b) = aE(X) + b. \tag{12}$$

Proof. Suppose that the distribution of X is continuous.

$$\begin{aligned} E[ag(X) + b] &= \int_{-\infty}^{\infty} [ag(X) + b] f(x) dx \\ &= \int_{-\infty}^{\infty} ag(X) f(x) dx + \int_{-\infty}^{\infty} b f(x) dx \\ &= aE[g(X)] + b. \end{aligned}$$

Expectation of Functions of a Random Variable

In general, $E[g(X)] \neq g(E(X))$.

Example

$$f(v) = \begin{cases} \dfrac{1}{a} & \text{for } 0 < v < a, \\ 0 & \text{otherwise.} \end{cases}$$

Let $W = g(V) = kV^2$.

Solution.

$$E(V) = \int_0^a v\frac{1}{a}dv = \frac{a}{2}.$$

$$g(E(V)) = k\left(\frac{a}{2}\right)^2 = \frac{1}{4}ka^2 \neq \frac{1}{3}ka^2 = E[g(V)].$$

In fact, unless g is a linear function, it is generally the case that $E[g(X)] \neq g(E(X))$.

Expectation of Functions of a Random Variable

Remark

If X is a random variable of mixed type, that is, a random variable that is discrete and continuous at the same time, then $E[g(X)]$ can be computed by decomposing the problem into two parts. For example, suppose that we toss a coin for which the probability of getting 隆掳tails隆卤is 3/8. If we get 隆掳tails,隆卤then the random variable X takes on the value 1; otherwise, X is a number taken at random in the interval $[2, 4]$. We can obtain $E[g(X)]$ as follows:

$$E[g(X)] = g(1) \cdot \frac{3}{8} + \Big(\int_2^4 g(x) \frac{1}{2} dx \Big) \cdot \frac{5}{8}.$$

Section 2.3 Mathematical Expectation and Variance

- 2.3.1 Expectation of a Random Variable
- 2.3.2 Expectation of Functions of a Random Variable
- 2.3.3 Variance of a Random Variable
- 2.3.4 The Application of Expectation and Variation

Variance of a Random Variable

The usefulness of the expected value as a prediction for the outcome of an experiment is increased when the outcome deviates too much from the expected value. In this section we shall introduce a measure, the variance, which is an indication of the variability of a random variable.

Definition

The ***variance*** *of a random variable X is defined by*

$$\sigma^2 \equiv E[(X - E[X])^2] = Var(X) = \begin{cases} \displaystyle\sum_{i=1}^{\infty} (x_i - \mu)^2 p(x_i) \\ \displaystyle\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \end{cases}$$

where $\mu = E(X)$.

Variance of a Random Variable

Remark

We deduce at once from the definition that the variance of X is always nonnegative (it can be infinite). Actually, it is strictly positive, except if the random variable X is a constant, which is a degenerate random variable. Finally, the larger the variance is, the more spread out is the distribution of the random variable around its mean.

Question

What does the variance mean?

Variance of a Random Variable

A first step in interpreting a new quantity is to figure out its units. The units of the variance are the squared units of X. This quantity is rather hard to interpret. To get back to units of X, we can take square root. The square root of the variance is the standard deviation, denoted by σ.

Definition

The ***standard deviation*** *of a random variable X is given by*

$$\sigma = SD(X) = \sqrt{Var(X)}.$$

Variance of a Random Variable

Property

$$Var(X) = E(X^2) - [E(X)]^2. \tag{13}$$

Proof. Assume X is discrete. Suppose that $\mu = E(X)$.

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] = E[(X - \mu)^2] \\ &= \sum_x (x - \mu)^2 p(x) = \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\ &= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \sum_x \mu^2 p(x) \\ &= E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2. \end{aligned}$$

The continuous case is similar to prove. □

Variance of a Random Variable

Example

Suppose that the p.d.f. of a random variable X is

$$f(x) = \begin{cases} \frac{x}{8} & \text{for } 0 < x < 4, \\ 0 & \text{otherwise,} \end{cases}$$

Find $Var(X)$.

Solution.

$$Var(X) = \int_0^4 \left(x - \frac{8}{3}\right)^2 \cdot \frac{1}{8} x dx = \int_0^4 x^2 \cdot \frac{1}{8} x dx - \left(\frac{8}{3}\right)^2 = \frac{8}{9}.$$

By using the equation (13), we also have

$$Var(X) = E(X^2) - E^2(X) = \int_0^4 x^2 \frac{1}{8} x dx - \left(\frac{8}{3}\right)^2 = \frac{8}{9}.$$

$\square$

Variance of a Random Variable

Theorem

$Var(X) = 0$ if and only if there exists a constant c such that $P(X = c) = 1$.

Proof. Suppose first that there exists a constant c such that $P(X = c) = 1$. Then $E(X) = c$, and $P[(X - c)^2 = 0] = 1$. Therefore,

$$Var(X) = E[(X - c)^2] = 0.$$

Conversely, suppose that $Var(X) = 0$. Then $P[(X - E(X))^2 \geqslant 0] = 1$ but $E[(X - E(X))^2] = 0$. Therefore, it can be seen that

$$P[(X - E(X))^2 = 0] = 1.$$

Hence, $P(X = E(X)) = 1$. $\square$

Variance of a Random Variable

Example

Suppose that X is a random variable for which $E(X) = \mu$ and $Var(X) = \sigma^2$. Show that $E[(X(X-1))] = \mu(\mu-1) + \sigma^2$.

Solution. For convenience, we assume that the distribution of X is continuous and the p.d.f. of X is f.

$$\begin{aligned}
E[(X(X-1))] &= \int_{-\infty}^{\infty} x(x-1)f(x)dx \\
&= \int_{-\infty}^{\infty} x^2 f(x)dx - \int_{-\infty}^{\infty} xf(x)dx \\
&= E(X^2) - E(X) = \{Var(X) + [E(X)]^2\} - E(X) \\
&= \sigma^2 + \mu^2 - \mu = \mu(\mu-1) + \sigma^2.
\end{aligned}$$

$\square$

Variance of a Random Variable

Property

For any constants a and b,

$$Var(ag(X) + b) = a^2 Var[g(X)].$$

In particular,

$$Var(aX + b) = a^2 Var(X).$$

Thus, the operator Var is not linear.

Proof. For any constants a and b,

$$\begin{aligned} Var(ag(X) + b) &= E\Big\{\big[(ag(X) + b) - E(ag(X) + b)\big]^2\Big\} \\ &= E\Big\{a^2\big[g(X) - E(g(X))\big]^2\Big\} = a^2 Var[g(X)]. \end{aligned}$$

$\square$

Variance of a Random Variable

Example

Let $X^* = \frac{X - E(X)}{\sqrt{Var(X)}}$. Find $E(X^*)$ and $Var(X^*)$.

Solution.

$$E(X^*) = E\Big(\frac{X - E(X)}{\sqrt{Var(X)}}\Big) = \frac{1}{\sqrt{Var(X)}}[E(X) - E(X)] = 0.$$

$$Var(X^*) = Var\Big(\frac{X - E(X)}{\sqrt{Var(X)}}\Big) = \frac{Var(X)}{Var(X)} = 1.$$

$\square$

Since $E(X^*) = 0$ and $Var(X^*) = 1$, X^* is usually said to be *standard random variable.*

Variance of a Random Variable

The mean and the variance of a random variable are particular cases of the quantities known as the ***moments*** of this variable.

Definition

The k***th moment about the origin****, or* ***noncentral moment*** *of a random variable* X *is defined by*

$$E(X^k) \equiv \mu_k^{'} = \begin{cases} \displaystyle\sum_{i=1}^{\infty} x_i^k p(x_i) & \textit{if } X \textit{ is discrete,} \\ \displaystyle\int_{-\infty}^{\infty} x^k f(x) dx & \textit{if } X \textit{ is continuous.} \end{cases}$$

Variance of a Random Variable

Definition

The k***th moment about the center****, or* ***central moment*** *of a random variable* X *is defined by*

$$E[(X-\mu)^k] \equiv \mu_k = \begin{cases} \sum_{i=1}^{\infty}(x_i-\mu)^k p(x_i) & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty}(x-\mu)^k f(x)dx & \text{if } X \text{ is continuous.} \end{cases}$$

We can easily find that
(i) $\mu_0^{'} = 1$ and $\mu_1^{'} = \mu = E(X)$.
(ii) $\mu_0 = 1, \mu_1 = 0$ and $\mu_2 = \sigma^2$.

Variance of a Random Variable

Theorem

If $E(|X|^k) < \infty$ for some positive integer k, then $E(|X|^j) < \infty$ for every integer j such that $j < k$.

Proof. For convenience, we assume that the distribution of X is continuous and the p.d.f. is f. Then

$$\begin{aligned} E(|X^j|) &= \int_{-\infty}^{\infty} |x^j| f(x) dx \\ &= \int_{|x| \leqslant 1} |x|^j f(x) dx + \int_{|x|>1} |x|^j f(x) dx \\ &\leqslant \int_{|x| \leqslant 1} 1 \cdot f(x) dx + \int_{|x|>1} |x|^j f(x) dx \\ &\leqslant P(|X| \leqslant 1) + E(|X|^k). \end{aligned}$$

By hypothesis, $E(|X|^k) < \infty$. It therefore follows that $E(|X|^j) < \infty$. $\square$

Variance of a Random Variable

Example

Suppose that X has a continuous distribution for which the p.d.f. has the following form

$$f(x) = ce^{-(x-3)^2}, \qquad -\infty < x < \infty.$$

Determine the mean of X and all the odd central moments.

Solution. It can be shown that for every positive integer k,

$$\int_{-\infty}^{\infty} |x|^k e^{-(x-3)^2} dx < \infty.$$

Variance of a Random Variable

Solution. Hence, all the moments of X exist.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} cxe^{-(x-3)^2}dx \\ &= \int_{-\infty}^{\infty} c(x-3)e^{-(x-3)^2}dx + 3\int_{-\infty}^{\infty} ce^{-(x-3)^2}dx \quad \text{let } t = x-3, \\ &= \int_{-\infty}^{\infty} cte^{-t^2}dt + 3 = \ 0 + 3 = \ 3. \end{aligned}$$

For every odd positive integer k, it follows that

$$E[(X-3)^k] = \int_{-\infty}^{\infty} c(x-3)^k e^{-(x-3)^2}dx = 0.$$

$\square$

Variance of a Random Variable

In the end of this section, we will introduce the concepts of the moment generating function and the characteristic function of a random variable X.

The ***moment generating function*** $M_X(t)$ of random variable X is defined by

$$M_X(t) = E(e^{tX}).$$

The ***characteristic function*** of a random variable X is defined as

$$\varphi_X(t) = E(e^{itX}), \qquad -\infty < t < \infty,$$

where $i = \sqrt{-1}$.

Section 2.3 Mathematical Expectation and Variance

- 2.3.1 Expectation of a Random Variable
- 2.3.2 Expectation of Functions of a Random Variable
- 2.3.3 Variance of a Random Variable
- 2.3.4 The Application of Expectation and Variation

The Application of Expectation and Variation

Property

(***Markov's inequality***) *If X is a random variable that takes only nonnegative values, then for any value $\varepsilon > 0$,*

$$P(X \geqslant \varepsilon) \leqslant \frac{E(X)}{\varepsilon}.$$

Proof. We give a proof for the case where X is continuous with density f.

$$\begin{aligned} E(X) &= \int_0^\infty x f(x) dx = \int_0^\varepsilon x f(x) dx + \int_\varepsilon^\infty x f(x) dx \\ &\geqslant \int_\varepsilon^\infty x f(x) dx \geqslant \int_\varepsilon^\infty \varepsilon f(x) dx = \varepsilon \int_\varepsilon^\infty f(x) dx = \varepsilon P(X \geqslant \varepsilon), \end{aligned}$$

and the result is proven. □

The Application of Expectation and Variation

Property

(***Chebyshev's inequality***) *If X is a random variable with mean μ and variance σ^2, then, for any value $\varepsilon > 0$,*

$$P(|X - \mu| \geqslant \varepsilon) \leqslant \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Since $(X - \mu)^2$ is a nonnegative random variable, we can apply Markov's inequality to obtain

$$P((X - \mu)^2 \geqslant \varepsilon^2) \leqslant \frac{E[(X - \mu)^2]}{\varepsilon^2}.$$

But since $(X - \mu)^2 \geqslant \varepsilon^2$ if and only if $|X - \mu| \geqslant \varepsilon$, the preceding is equivalent to

$$P(|X - \mu| \geqslant \varepsilon) \leqslant \frac{E[(X - \mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}$$

and the proof is complete. $\square$

The Application of Expectation and Variation

From Markov's inequality, we can see if $E(X)$ is small, then it is not too likely that X is large.

Chebyshev's inequality tells us that if $\sigma^2 = Var(X)$ is small, then it is not too likely that X is far from its mean.

The importance of Markov's and Chebyshev's inequalities is that they enable us to derive bounds on probabilities when only the mean, or both the mean and the variance, of the probability distribution are known.

The Application of Expectation and Variation

Example

Suppose that we know that the number of items produced in a factory during a week is a random variable with mean 500.

(a) What can be said about the probability that this week's production will be at least 1000?

(b) If the variance of a week's production is known to equal 100, then what can be said about the probability that this week's production will be between 400 and 600?

The Application of Expectation and Variation

Solution. Let X be the number of items that will be produced in a week.
(a) By Markov's inequality,

$$P(X \geqslant 1000) \leqslant \frac{E(X)}{1000} = \frac{500}{1000} = \frac{1}{2}.$$

(b) By Chebyshev's inequality,

$$P(|X - 500| \geqslant 100) \leqslant \frac{\sigma^2}{(100)^2} = \frac{1}{100}.$$

Hence,

$$P(|X - 500| \leqslant 100) \geqslant 1 - \frac{1}{100} = \frac{99}{100},$$

and so the probability that this week's production will be between 400 and 600 is at least 0.99. $\square$

The Application of Expectation and Variation

Example

Let X be any random variable with $E(X) = \mu$ and $Var(X) = \sigma^2$. Then, Chebyshev's inequality states that

$$P(|X - \mu| \geqslant k\sigma) \leqslant \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.$$

Thus, for any random variable, the probability of a deviation from the mean of more than k standard deviations is less than or equal to $\frac{1}{k^2}$. $\square$

The Application of Expectation and Variation

Chebyshev's inequality is the best possible inequality in the sense that, for any $\varepsilon > 0$, it is possible to give an example of a random variable for which Chebyshev's inequality is in fact an equality. To see this, given $\varepsilon > 0$, choose X with distribution

$$p_X(-\varepsilon) = \frac{1}{2}, \quad p_X(\varepsilon) = \frac{1}{2}.$$

Then $E(X) = 0, Var(X) = \varepsilon^2$, and

$$P(|X - \mu| \geqslant \varepsilon) = \frac{Var(X)}{\varepsilon^2} = 1.$$

Section 2.4 Discrete Random Variables

- 2.4.1 Binomial Distribution with Parameters n and p
- 2.4.2 Geometric Distribution
- 2.4.3 Poisson Distribution with Parameters λ

Binomial Distribution with Parameters n and p

It would be very tedious if, every time we had a slightly different problem, we had to determine the probability distributions from scratch. Luckily, there are enough similarities between certain types, or families, of experiments, to make it possible to develop formulas representing their general characteristics.

Binomial Distribution with Parameters n and p

we have known that a Bernoulli trial is a random experiment in which there are only two possible outcomes: success and failure. Now we establish a random variable X which assigns the value 1 to the outcome "success" and assigns the value 0 to the outcome "failure". Suppose that

$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad 0 < p < 1.$$

We call X a ***Bernoulli random variable***. We can say X has ***Bernoulli distribution with parameter*** p. The p.f. of X is written as follows:

$$p(x) = \begin{cases} 1 - p & \text{if } x = 0, \\ p & \text{if } x = 1. \end{cases}$$

Binomial Distribution with Parameters n and p

It can be easily checked that the mean and variance of X are

$$E(X) = p, \qquad Var(X) = p(1-p).$$

Binomial Distribution with Parameters n and p

Consider a sequence of Bernoulli trials, i.e.,

(i) the experiment consists of n repeated Bernoulli trials - each trial has only two possible outcomes labeled as success and failure,

(ii) the trials are independent - the outcome of any trial has no effect on the probability of the others,

(iii) the probability of success in each trial is constant which we denote by p.
Similar to establish the Binomial random variable, we can establish

Binomial Distribution with Parameters n and p

Definition

The random variable X that counts the number of successes, k, in the n Bernoulli trials is said to have a ***binomial distribution with parameters*** n ***and*** p*, written as $X \sim B(n, p)$.*

The probability function of the random variable $X \sim B(n, p)$ is given by

$$p(x) = \binom{n}{x} p^x q^{n-x} \quad \text{for } x = 0, 1, \cdots, n, \tag{14}$$

where $q = 1 - p$.

Binomial Distribution with Parameters n and p

A parameter is a symbol that appears in the definition of a random variable and that can take different values. For example, in the case of the binomial distribution, n can take on the values $1, 2, \cdots$, and p all the values in the interval $(0, 1)$. In practice, the parameter p is generally unknown.

If $p = 0$ or 1, we can get ***degenerate distribution***. The probability function becomes $P(X = 1) = 1$. In fact, it can be generated described by

$$P(X = C) = 1, \qquad \text{where } C \text{ is a constant.}$$

But for simplicity, we still think of it as a random variable.

Binomial Distribution with Parameters n and p

Remark

(i) *We have that* $p(x) \geqslant 0$ *for all* x. *By Newton隆炉s binomial formula,*

$$\sum_{x=0}^{n} p(x) = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x} = (p+q)^n = 1,$$

as should be.

(ii) *The shape and the position of the probability function* $p(x)$ *depend on the parameters* n *and* p. *When* $p = 0.5$, *the binomial distribution is symmetrical.*

Binomial Distribution with Parameters n and p

Property

Suppose that X is a random variable which has binomial distribution with parameters n and p. Then we have

$$E(X) = np \quad \text{and} \quad Var(X) = np(1-p). \tag{15}$$

Binomial Distribution with Parameters n and p

Proof.

$$
\begin{aligned}
E(X) &= \sum_{x=0}^{n} xp(x) = \sum_{x=0}^{n} x \cdot \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=0}^{n} x \cdot \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\
&= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\
&= np \sum_{x=1}^{n} \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x} \\
&= np[p + (1-p)]^n - 1 = np.
\end{aligned}
$$

Binomial Distribution with Parameters n and p

Proof.

$$
\begin{aligned}
& Var(X) \\
&= E(X^2) - [E(X)]^2 \\
&= \sum_{x=0}^{n} x^2 p(x) - (np)^2 = \sum_{x=0}^{n} x^2 \cdot \binom{n}{x} p^x (1-p)^{n-x} - (np)^2 \\
&= \sum_{x=0}^{n} \cdot \frac{x(x-1)n!}{x!(n-x)!} p^x (1-p)^{n-x} + \sum_{x=0}^{n} x \cdot \binom{n}{x} p^x (1-p)^{n-x} - (np)^2 \\
&= n(n-1)p^2 \sum_{x=2}^{n} \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} (1-p)^{(n-2)-(x-2)} + np - (np)^2 \\
&= n(n-1)p^2 \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} (1-p)^{n-x} + np - (np)^2 \\
&= n(n-1)p^2 + np - (np)^2 = np(1-p).
\end{aligned}
$$

$\square$

Binomial Distribution with Parameters n and p

We can also think about $E(X)$ from another aspect. Since p is the probability of success in each trial, it is reasonable to regard p as the number of success in one trial. So if we repeat the experiment n times independently, the number of success should be np. This will be proved in next chapter.

Binomial Distribution with Parameters n and p

Example

In an airport, five radars are in operation and each radar has a 0.9 probability of detecting an arriving airplane. The radars operate independently of each other.

(a) Calculate the probability that an arriving airplane will be detected by at least four radars.

(b) Knowing that at least three radars detected a given airplane, what is the probability that the five radars detected this airplane?

(c) What is the smallest number n of radars that must be installed if we want an arriving airplane to be detected by at least one radar with probability 0.9999?

Binomial Distribution with Parameters n and p

Solution. Let X be the number of radars that detect the airplane.
(a) We have that $X \sim B(5,\ 0.9)$. We find Table of Binomial Probabilities in Appendix, that

$$P(X \geqslant 4) = \binom{5}{4}(0.9)^4(0.1) + (0.9)^5 = 0.5905 + 0.3280 = 0.9185.$$

(b) We want

$$\begin{aligned} P(X = 5|X \geqslant 3) &= \frac{P(\{X = 5\} \cap \{X \geqslant 3\})}{P(\{X \geqslant 3\})} = \frac{P(X = 5)}{P(X \geqslant 3)} \\ &= \frac{0.5905}{0.5905 + 0.3280 + 0.0729} \approx 0.596. \end{aligned}$$

Binomial Distribution with Parameters n and p

Solution. (c)We now have that $X \sim B(n,\ 0.9)$ and we seek (the smallest) n such that $P(X \geqslant 1) = 0.9999$. We have:

$$\begin{aligned} P(X \geqslant 1) =& 1 - P(X = 0) = 1 - \binom{n}{0}(0.9)^0(0.1)^n = 1 - (0.1)^n \\ =& 0.9999 \Longleftrightarrow (0.1)^n = 0.0001 = (0.1)^4. \end{aligned}$$

Thus, we may write that $n_{min} = 4$. $\square$

Binomial Distribution with Parameters n and p

Example

A quality control engineer is in charge of testing whether or not 90% of the DVD players produced by his company conform to specifications. To do this, the engineer randomly selects a batch of 12 DVD players from each day's production. The day's production is acceptable provided no more than 1 DVD player fails to meet specifications. Otherwise, the entire day's production has to be tested.

(a) What is the probability that the engineer incorrectly passes a day's production as acceptable if only 80% of the day's DVD players actually conform to specification?

(b) What is the probability that the engineer unnecessarily requires the entire day's production to be tested if in fact 90% of the DVD players conform to specifications?

Binomial Distribution with Parameters n and p

Solution. Let X denote the number of DVD players in the sample that fail to meet specifications. In part (a) we want $P(X \leqslant 1)$ with binomial parameters $n = 12$, $p = 0.2$.

$$\begin{aligned} P(X \leqslant 1) &= P(X = 0) + P(X = 1) \\ &= \binom{12}{0} 0.2^0 0.8^{12} + \binom{12}{1} 0.2^1 0.8^{11} = 0.069 + 0.206 = 0.275 \end{aligned}$$

(b) We now want $P(X > 1)$ with binomial parameters $n = 12$, $p = 0.1$.

$$\begin{aligned} P(X \leqslant 1) &= P(X = 0) + P(X = 1) \\ &= \binom{12}{0} 0.1^0 0.9^{12} + \binom{12}{1} 0.1^1 0.9^{11} = 0.069 + 0.206 = 0.659 \end{aligned}$$

So $P(X > 1) = 0.341$. $\square$

Binomial Distribution with Parameters n and p

Example

Suppose that a baseball player's batting average is 0.33 (1 for 3 on average). Consider the case where the player either gets a hit or makes an out (forget about walks here!). On average how many hits does the player get in 100 at bats? What's the standard deviation for the number of hits in 100 at bats?

Solution. It is obvious that the probability for a hit is $p = 0.33$ and the probability for 隆掳no hit隆卤is $q = 1 - p = 0.67$. Let X be the hit number. Then $X \sim B(100, 0.33)$.

$$\mu = np = 100 \cdot 0.33 = 33 \text{ hits} \qquad \text{and}$$

$$\sigma = \sqrt{npq} = \sqrt{100 \cdot 0.33 \cdot 0.67} \approx 4.7 \text{ hits.}$$

□

Binomial Distribution with Parameters n and p

Example

Given that 5% of a population are left-handed, calculate the probability that a random sample of 100 people contains 2 or more left-handed people.

Solution. Let X be the number of left handed people in a sample of 100. Then $X \sim B(100, 0.05)$. So

$$P(X = x) = \binom{100}{x} 0.05^x 0.95^{100-x}, \quad x = 0, 1, \cdots, 100$$

What we want is $P(X \geqslant 2)$.

$$\begin{aligned} P(X \geqslant 2) &= 1 - P(X < 2) \\ &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - \binom{100}{0} 0.95^{100} - \binom{100}{1} 0.05 \cdot 0.95^{99}. \end{aligned}$$

$\square$

Binomial Distribution with Parameters n and p

The calculation will make us crazy if we have no computer. The Poisson limit theorem or law of rare events gives a Poisson approximation to the binomial distribution, under certain conditions. The theorem was named after Siméon Denis Poisson (1781-1840).

Theorem

(***Poisson limit theorem***) *Suppose that λ is a constant and n is a positive integer. If $\lim_{n\to\infty} np_n = \lambda$, then we have*

$$\lim_{n\to+\infty} \binom{n}{x} p_n^x (1-p_n)^{n-x} = e^{-\lambda}\frac{\lambda^x}{x!}$$

for any fixed nonnegative integer x.

Binomial Distribution with Parameters n and p

Proof. We rewrite $p_n = \lambda/n$ $(p \to +\infty)$. Now we have

$$
\begin{aligned}
&\lim_{n\to+\infty} \binom{n}{x}\left(\frac{\lambda}{n}\right)^x\left(1-\frac{\lambda}{n}\right)^{n-x} \\
=&\lim_{n\to+\infty} \frac{n!}{x!(n-x)!}\frac{\lambda^x}{n^x}\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-x} \\
=&\lim_{n\to+\infty} \frac{n(n-1)\cdots(n-x+1)}{x!}\frac{\lambda^x}{n^x}\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-x} \\
=&\lim_{n\to+\infty} \frac{\lambda^x}{x!}\frac{n(n-1)\cdots(n-x+1)}{n\cdot n\cdots n}\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-x} \\
=&\lim_{n\to+\infty} \frac{\lambda^x}{x!}\left[1\cdot\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{x-1}{n}\right)\right]\left(1-\frac{\lambda}{n}\right)^n\left(1-\frac{\lambda}{n}\right)^{-x} \\
=&\frac{\lambda^x}{x!}\cdot 1\cdot e^{-\lambda}\cdot 1 = \frac{e^{-\lambda}\lambda^x}{x!}.
\end{aligned}
$$

$\square$

Binomial Distribution with Parameters n and p

In general, if n is large (say > 50, (> 20)) and p is small (say < 0.1 (< 0.05)), then we can use Poisson limit theorem. Now by using Poisson approximation, we can get $\lambda = 100 \cdot 0.05 = 5$, and

$$\begin{aligned} P(X \geqslant 2) &= 1 - \binom{100}{0} 0.95^{100} - \binom{100}{1} 0.05 \cdot 0.95^{99} \\ &\approx 1 - e^{-5}\frac{5^0}{0!} - e^{-5}\frac{5^1}{1!} \\ &\approx 1 - 0.0067 - 0.0337 \approx \ 0.9596. \end{aligned}$$

Above example tells us that the impossible principle of small probability events, while this example wants to tell us people shouldn't neglect small probability because these events are almost inevitable with the experiment repeated again and again.

Section 2.4 Discrete Random Variables

- 2.4.1 Binomial Distribution with Parameters n and p
- 2.4.2 Geometric Distribution
- 2.4.3 Poisson Distribution with Parameters λ

Geometric Distribution

Example

A representative from the National Football League's Marketing Division randomly selects people on a random street in Kansas City, Kansas until he finds a person who attended the last home football game. Let p, the probability that he succeeds in finding such a person, equal 0.20. And, let X denote the number of people he selects until he finds his first success.

What is the probability function of X?

Geometric Distribution

Example

It is obvious that

$$p = p(\text{success}) = 0.2 \text{ and } 1 - p = p(\text{failure}) = 0.8.$$

Let X denote the number of people he selects until he finds his first success.

$$\{X = x\} \text{ means } \underbrace{FF\cdots F}_{(x-1)\text{ failures}} \underbrace{S}_{1\text{ success}},$$

so $P(X = x) = P(F)^{x-1}P(S) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, \cdots.$

$\square$

Geometric Distribution

Definition

Let X be the random variable that counts the number of Bernoulli trials needed to obtain a first success. We say that X has a ***geometric distribution with parameter*** *p, where p is the probability of success on any trial. We write that $X \sim G(p)$ (or $Geom(p)$). The p.f. of X is*

$$p(x) = q^{x-1}p$$

for $x = 1, 2, \cdots$, where $q = 1 - p$.

Geometric Distribution

Remark

We have that $p(x) \geqslant 0$ *for all* x *and*

$$\sum_{x=1}^{\infty} p(x) = p \sum_{x=1}^{\infty} q^{x-1} = p \frac{1}{1-q} = 1.$$

Thus the probability function $p(x)$ *is a valid probability function.*

Geometric Distribution

The distribution function of X is given by

$$F(x) = P(X \leqslant x) = \sum_{k=1}^{x} p(k) = p \sum_{k=1}^{x} q^{k-1} = p\frac{1-q^x}{1-q} = 1 - q^x$$

for $x = 1, 2, \cdots$. Then we deduce that $P(X > x) = q^x$. Making use of the formula, we can show that

$$P(X > x + y | X > x) = P(X > y)$$

for any $x, y \in 0, 1, 2, \cdots$. This property is known as the ***memoryless property*** of the geometric distribution.

Geometric Distribution

Property

Suppose that X is a random variable which has a geometric distribution with parameters p. Then we have

$$E(X) = \frac{1}{p} \quad \text{and} \quad Var(X) = \frac{1-p}{p^2}. \tag{16}$$

Proof.

$$\begin{aligned} E(X) &= \sum_{x=1}^{\infty} xp(x) = \sum_{x=1}^{+\infty} x \cdot q^{x-1} p \\ &= p \sum_{x=1}^{+\infty} (q^x)' = p\Big(\sum_{x=1}^{+\infty} q^x\Big)' = p\Big(\frac{q}{1-q}\Big)' = \frac{1}{p}. \end{aligned}$$

Geometric Distribution

Proof.

$$
\begin{aligned}
Var(X) &= E(X^2) - [E(X)]^2 \\
&= \sum_{x=1}^{+\infty} x^2 p(x) - \frac{1}{p^2} = \sum_{x=1}^{+\infty} x^2 \cdot q^{x-1} p - \frac{1}{p^2} \\
&= \sum_{x=1}^{+\infty} x(x-1) \cdot q^{x-1} p + \sum_{x=1}^{+\infty} x p(x) - \frac{1}{p^2} \\
&= pq \sum_{x=1}^{+\infty} (q^x)'' + \frac{1}{p} - \frac{1}{p^2} = pq \Big(\frac{q}{1-q} \Big)'' + \frac{1}{p} - \frac{1}{p^2} \\
&= pq \cdot \frac{2}{(1-q)^3} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}.
\end{aligned}
$$

$\square$

Geometric Distribution

In fact, we can calculate the value $E(X)$ by another way. Consider the following triangular array of probabilities:

$$\begin{array}{cccc} P(X=1) & P(X=2) & P(X=3) & \cdots \\ & P(X=2) & P(X=3) & \cdots \\ & & P(X=3) & \cdots \\ & & & \cdots \end{array}$$

The above way to calculate the sum of all the elements is adding the elements in each column of the array and then adding these column totals. Now we can add the elements in each row of the array and then add these row totals. In this way we obtain the value $\sum_{x=1}^{\infty} P(X \geqslant x)$. Theorefore

$$E(X) = \sum_{x=1}^{\infty} P(X \geqslant x) = \sum_{x=1}^{\infty} (1-p)^{x-1} = \frac{1}{p}.$$

Geometric Distribution

Example

A man shoots at a target until he has hit it twice. Suppose that the probability that a given shot hits the target is equal to 0.8. If the man stops shooting as soon as he hits the target, what is the probability that the man must shoot exactly four times?

Solution. Let X be the number of shots needed to end the random experiment. Then, if we assume that the shots are independent, we may write that $X \sim G(0.8)$ and

$$P(X = 4) = (1 - 0.8)^3(0.8) = 0.0064.$$

$\square$

Geometric Distribution

Example

Watch the input queue at the alpha farm for a job that times out. The probability that a job times out is 0.05. Let Y be the number of the first job to time out, then $Y \sim G(0.05)$. What's then the probability that
(a) the third job times out?
(b) Y is less than 3?
(c) the first job to time out is between the third and the seventh?

Solution. (a) $P(Y = 3) = 0.95^2(0.05) = 0.045$.
(b) $P(Y < 3) = P(Y \leqslant 2) = 1 - 0.95^2 = 0.0975$.
(c) $P(3 \leqslant Y \leqslant 7) = P(Y \leqslant 7) - P(Y \leqslant 2) =$
$1 - 0.95^7 - (1 - 0.95^2) = 0.204$. $\square$

Section 2.4 Discrete Random Variables

- 2.4.1 Binomial Distribution with Parameters n and p
- 2.4.2 Geometric Distribution
- 2.4.3 Poisson Distribution with Parameters λ

Poisson Distribution with Parameters λ

Let us see some examples.

(i) Let X be the number of typos on a printed page.
(ii) Let X be the number of cars passing through an intersection in one minute.
(iii) Let X be the number of Alaskan salmon caught in a squid driftnet.
(iv) Let X be the number of customers at an ATM in 10-minute intervals.
(v) Let X be the number of students arriving during office hours.

These kind of random variables has a common characteristic: they all denote the number of times an event occurs in an interval of time (or space). We call them Poisson random variables.

Poisson Distribution with Parameters λ

In fact, let X be a random variable having a binomial distribution with parameters n and p. We can show that if n tends to infinity and p decreases to 0, in such a way that the product np remains equal to the constant λ, then the probability function of X converges to the function $p(x)$ given by

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!} \text{ for } x = 0, 1, \cdots . \tag{17}$$

Definition

We say that the discrete random variable X whose probability function is given by equation (17) has a ***Poisson distribution with parameter*** $\lambda > 0$. *We write that $X \sim P(\lambda)$.*

Poisson Distribution with Parameters λ

Remark

(i) *Making use of the formula*

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots ,$$

we easily show that the function defined in (17) is a valid probability function.

(ii) *The shape of the probability function $p(x)$ depends on the value of the parameter λ. We observe that the distributions are centered roughly on λ, exhibit positive skew (that decreases a λ increases) and the variance (spread) increases as λ increases.*

(iii) *To evaluate the distribution function of a Poisson random variable, we can use a pocket calculator or a statistical software package. Table of Poisson Probabilities in Appendix gives many values of this function.*

Poisson Distribution with Parameters λ

Remark

(iv) *We deduce from what precedes that we can use a Poisson distribution with parameter* $\lambda = np$ *to approximate the binomial distribution with parameters* n *and* p. *In general, the Poisson approximation should be good if* $n > 20$ *and* $p < 0.05$. *If the value of* p *is greater than 1/2, then we must consider the number of failures (with probability* $1 - p < 1/2$*) rather than the number of successes before performing the approximation by the Poisson distribution.*

Poisson Distribution with Parameters λ

Property

Suppose that X is a random variable which has Poisson distribution with parameter λ. Then we have

$$E(X) = Var(X) = \lambda. \tag{18}$$

Proof.

$$\begin{aligned} E(X) = &\sum_{x=0}^{n} xp(x) = \sum_{x=0}^{+\infty} x \cdot \frac{e^{-\lambda}\lambda^{x}}{x!} \\ =&\lambda \sum_{x=1}^{+\infty} \frac{e^{-\lambda}\lambda^{x-1}}{(x-1)!} = \lambda \sum_{x'=0}^{+\infty} \frac{e^{-\lambda}\lambda^{x'}}{x'!} = \lambda. \end{aligned}$$

Poisson Distribution with Parameters λ

Proof.

$$
\begin{aligned}
Var(X) = E(X^2) - [E(X)]^2 &= \sum_{x=0}^{+\infty} x^2 p(x) - \lambda^2 \\
&= \sum_{x=0}^{+\infty} x(x-1) \cdot p(x) + \sum_{x=1}^{+\infty} x p(x) - \lambda^2 = \lambda.
\end{aligned}
$$

$\square$

Poisson Distribution with Parameters λ

Example

Births in a hospital occur randomly at an average rate of 1.5 births per hour. What is the probability of observing 4 births in a given hour at the hospital?

Solution. Let X be the number of births in a given hour. Since the events occur randomly and mean rate is $\lambda = 1.5$, we assume $X \sim P(1.5)$. We can now use the equation (17) to calculate the probability of observing exactly 4 births in a given hour.

$$P(X = 4) = e^{-1.5}\frac{1.5^4}{4!} = 0.0471.$$

$\square$

Poisson Distribution with Parameters λ

Example

Suppose that in a large population the proportion of people that have a certain disease is 0.01. Determine the probability that in a random group of 200 people at least four people will have the disease.

Solution. Let X be the number of people having the disease among the 200 people in the random group. It is obvious that $X \sim B(200,\ 0.01)$. This distribution can be approximated by a Poisson distribution for which the mean is $\lambda = np = 2$. Let Y denote a random variable having this Poisson distribution, i.e., $Y \sim P(2)$. Then it can be found from the Table of the Poisson probabilities in Appendix at the end of this book that

Poisson Distribution with Parameters λ

Solution.

$$\begin{aligned} & P(X \geqslant 4) \approx P(Y \geqslant 4) \\ &= 1 - P(Y = 0) - P(Y = 1) - P(Y = 2) - P(Y = 3) \\ &= 1 - 0.1353 - 0.2707 - 0.2707 - 0.1805 = 0.1428. \end{aligned}$$

Hence, the probability that at least four people will have the disease is approximately 0.1428. The actual value is 0.1420. $\square$

The idea of using one distribution to approximate another is widespread throughout statistics and one we will meet again. In many situations it is extremely difficult to use the exact distribution and so approximations are very useful.

Poisson Distribution with Parameters λ

Example

A new type of brakes is being studied. The company manufacturing these brakes claims that they could last at least 100,000 km for 90% of the vehicles that will use them. A laboratory simulated the driving of 100 cars using these brakes. Let X be the number of cars whose brakes will not last 100,000 km.

(a) What distribution does X follow?

(b) We will doubt the claimed percentage if the brakes must be changed on 17 cars or more before 100,000 km. What is, approximately, the probability of observing this event if, in fact, the claimed percentage is exact?

Poisson Distribution with Parameters λ

Solution. (a) By definition, if we assume that the cars are independent, then we may write that $X \sim B(100, 0.1)$.
(b) We want $P(X \geqslant 17)$. We have that
$P(X \geqslant 17) \approx P(Y \geqslant 17)$, where $Y \sim P(\lambda)$, $\lambda = 100 \cdot 0.1 = 10$.
We then find in Table of Poisson Probabilities, that
$P(Y \geqslant 17) \approx 0.0271$.

Poisson Distribution with Parameters λ

Remark

Here, we have that $n = 100$ and $p = 0.1$. The value of n is very large, which is preferable, but that of p is slightly too large to expect a good approximation. Actually, we find that $P(X \geqslant 17) \approx 0.021$.

Poisson Distribution with Parameters λ

The Poisson distribution is named after Simeon-Denis Poisson (1781-1840). It was first described in Researches on the probability of criminal and civil verdicts. The goal of the text is to analyze the relationship between the likelihood of conviction of the accused and the likelihood of the individual's actually having committed the crime. (Estimates of this type enable one to determine approximately how many innocent people are locked away in jail. Unfortunately, they give no insight into which people are innocent.) It was during the course of his analysis that Poisson briefly described a new kind of probability curve or distribution.

Poisson Distribution with Parameters λ

Poisson's distribution enables the user to calculate the likelihood that a certain event will occur x times in a given time interval, where x represents any whole number greater than or equal to 0. This discovery passed without much notice during Poisson's time. Perhaps the reason it did not draw much attention was that he could not find an eye-catching application for his insight, but conditions have changed. Poisson distribution is now widely used.

Poisson Distribution with Parameters λ

For example, it is employed when developing probabilistic models of telephone networks, in which they are used to predict the probability that x phone calls will arrive at a particular point on the network in a given interval of time. They are also used in the design of traffic networks in a similar sort of way. (Car arrival times are studied instead of message arrival times.) Neither of these applications could have been foreseen by Poisson or his contemporaries, of course.

Section 2.5 Continuous Random Variables

- 2.5.1 Uniform Distribution
- 2.5.2 Exponential Distribution
- 2.5.3 Normal Distribution

Continuous Random Variables

There are a lot of models which can not be described by discrete random variables. For example, if one measures the width of an oak leaf, the results may take values between 3cm and 4cm at random. If we observe the lifetime of a car, the results may change continuously between 0 year and 30 years.

In this section, we will introduce three important continuous random variables.

Uniform Distribution

Let a and b be two given real numbers such that $a < b$.

Definition

The distribution of the random variable X is called the ***uniform distribution*** *of the interval $[a, b]$ if the p.d.f. of X is*

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \leqslant x \leqslant b, \\ 0 & \text{otherwise.} \end{cases}$$

We write that by $X \sim U(a, b)$.

The corresponding d.f. of X is

$$F(x) = \begin{cases} 0 & \text{for } x < a, \\ \dfrac{x-a}{b-a} & \text{for } a \leqslant x < b, \\ 1 & \text{for } x \geqslant b. \end{cases}$$

Uniform Distribution

The constant a is the *location parameter* and the constant $b - a$ is the *scale parameter*.

The case where $a = 0$ and $b = 1$ is called the ***standard uniform distribution***. The p.d.f. for the standard uniform distribution is

$$f(x) = \begin{cases} 1 & \text{for } 0 \leqslant x \leqslant 1, \\ \\ 0 & \text{otherwise.} \end{cases}$$

and the d.f. of the standard uniform distribution is

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ x & \text{for } 0 \leqslant x < 1, \\ 1 & \text{for } x \geqslant 1. \end{cases}$$

Uniform Distribution

In fact, uniform distribution is coming from the following probability model.

Consider an experiment which a point X is chosen at random from the interval $\Omega = \{x : a < x < b\}$ in such a way that the probability that X will belong to each subinterval of Ω is proportional to the length of that subinterval.

If X has a uniform distribution in an interval Ω, then X must take a value in the interval Ω. Hence, the p.d.f. $f(x)$ of X must be 0 outside of Ω. Furthermore, since any particular subinterval of Ω having a given length is as likely to contain X as is any other subinterval having the same length, regardless of the location of the particular subinterval in Ω, it follows that $f(x)$ must be a constant throughout Ω.

Uniform Distribution

We also have

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{a}^{b} f(x)dx = 1. \tag{19}$$

Therefore, the constant value of $f(x)$ throughout Ω must be $1/(b-a)$.

It is seen from equation (19) that the p.d.f. representing a uniform distribution in a given interval is constant over that interval, and the constant value of the p.d.f. is the reciprocal of the length of the interval. It is not possible to define a uniform distribution over the interval $X \geqslant a$ because the length of this interval is infinite.

Uniform Distribution

For the uniform distribution in the interval $[a, b]$, since the probability is 0 that one of the endpoints a or b will be chosen, it is irrelevant whether the distribution is regarded as a uniform distribution in the closed interval $a \leqslant x \leqslant b$, or as a uniform distribution in the open interval $a < x < b$, or as a uniform distribution in the half-open and half-closed interval $(a, b]$ in which one endpoint is included and the other endpoint is excluded.

Uniform Distribution

Property

Suppose that X is a random variable which has uniform distribution of the interval $[a, b]$. Then we have

$$E(X) = \frac{a+b}{2} \quad \text{and} \quad Var(X) = \frac{(b-a)^2}{12}. \tag{20}$$

Proof. Using the basic definition of expectation, we know

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}.$$

We have

$$\begin{aligned} Var(X) =& E(X^2) - [E(X)]^2 \\ =& \int_a^b x^2 \cdot \frac{1}{b-a} dx - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}. \end{aligned}$$

□

Uniform Distribution

Example

The current (in mA) measured in a piece of copper wire is known to follow a uniform distribution over the interval [0, 25]. Write down the formula for the probability density function $f(x)$ of the random variable X representing the current. Calculate the expectation and variance of the distribution and find the distribution function $F(x)$.

Solution. Over the interval [0, 25] the probability density function $f(x)$ is given by the formula

$$f(x) = \begin{cases} \dfrac{1}{25-0} = 0.04 & \text{for } 0 \leqslant x \leqslant 25, \\ 0 & \text{otherwise.} \end{cases}$$

Using equation (20), we have

Uniform Distribution

Solution. Using equation (20), we have

$$E(X) = \frac{25+0}{2} = 12.5mA \text{ and } Var(X) = \frac{(25-0)^2}{12} = 52.08mA^2.$$

The distribution function is obtained by integrating the probability density function as shown below,

$$F(x) = \int_{-\infty}^{x} f(t)dt.$$

Hence, choosing the three distinct regions $x < 0$, $0 \leqslant x < 25$ and $x \geqslant 25$ in turn gives:

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ \dfrac{x}{25} & \text{for } 0 \leqslant x < 25, \\ 1 & \text{for } x \geqslant 25. \end{cases}$$

□

Uniform Distribution

Example

Consider the experiment of choosing a point at random from the disk A of radius R in the plane centered at the origin. To make the experiment more interesting, we can think of it as the result of throwing a dart at a disk-shaped target. Let X be the random variable denoting the distance of the point chosen from the origin. Find the distribution function $F(x)$ and calculate the probability $P(a < X \leqslant b)$, where $0 \leqslant a \leqslant b \leqslant R$.

Solution. We can say that $X \sim U(A)$. If $0 \leqslant x \leqslant R$, the event $\{\omega | X(\omega) \leqslant x\}$ is the disk in the plane, of radius x centered at the origin. Its area is πx^2. Thus

Uniform Distribution

Solution.

$$P(X \leqslant x) = \frac{\pi x^2}{\pi R^2} = \frac{x^2}{R^2}, \quad 0 \leqslant x \leqslant R.$$

If $x < 0$, then $P(X \leqslant x) = 0$. If $x > R$, then $P(X \leqslant x) = 1$. Thus the distribution of X is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ x^2/R^2 & \text{for } 0 \leqslant x \leqslant R, \\ 1 & \text{for } x > R. \end{cases}$$

Furthermore, if $0 \leqslant a \leqslant b$, then

$$P(a < X \leqslant b) = F(b) - F(a) = \frac{b^2 - a^2}{R^2}.$$

$\square$

Uniform Distribution

Example

Given a random variable X with distribution $F_X(x)$ which is strict increasing, prove that the function $g(x)$ such that the random variable $Y = g(X)$ is uniformly distributed in the interval (0, 1) must be $F_X(x)$.

(*Hint: if* $Y = g(X) = F_X(X)$, *then* $F_Y(y) = y$ for $0 \leqslant y \leqslant 1$.)

Proof. Suppose that x is an arbitrary number and $y = g(x) = F_X(x)$. By using the monotonicity of $F_X(x)$, for the random variable $Y = F_X(X)$, we get if $y < 0$, then

Uniform Distribution

Proof.

$$F_Y(y) = P(Y \leqslant y) = P(F_X(X) \leqslant y) < P(F_X(X) \leqslant 0) = 0.$$

If $y > 1$, then

$$1 \geqslant F_Y(y) = P(Y \leqslant y) = P(F_X(X) \leqslant y) \geqslant P(F_X(X) \leqslant 1) = 1.$$

So $F_Y(y) = 1$ if $y > 1$. If $0 \leqslant y \leqslant 1$, then

$$F_Y(y) = P(Y \leqslant y) = P(F_X(X) \leqslant y) = P(X \leqslant F_X^{-1}(y)) = F\big(F_X^{-1}(y)\big)$$

That is

$$F_Y(y) = \begin{cases} 0 & \text{for } y < 0, \\ y & \text{for } 0 \leqslant y \leqslant 1, \\ 1 & \text{for } y > 1. \end{cases}$$

i.e., $X \sim U(0,1)$. $\square$

Uniform Distribution

Example

Given a random variable Y with uniform distribution of the interval (0, 1), find a function $h(y)$ such that the distribution of the random variable $X = h(Y)$ is a specified function $F_X(x)$.

Solution. Let $h(y)$ be the inverse of the function $F_X(x)$, i.e., $x = h(y) = F_X^{-1}(y)$. For the random variable $X = F_X^{-1}(Y)$ and $x \in \mathbb{R}$,

$$P(X \leqslant x) = P(F_X^{-1}(Y) \leqslant x) = P(Y \leqslant F_X(x)) = F_X(x).$$

$\square$

Uniform Distribution

Remark: The random variable Y can be considered as the output of a nonlinear memoryless system with input X and transfer characteristic $F_X(x)$. Therefore if we use Y as the input to another system with transfer characteristic the inverse $F_X^{-1}(y)$ of the function $y = F_X(x)$, the resulting output will equal X which distribution is $F_X(x)$ if we take $X = F_X^{-1}(y)$.

Uniform Distribution

Uniform distribution are always used to produce random digits. Virtually since the idea arose to maintain a storehouse of random digits, scientists have been coming up with more and better ways of using them. Today, random numbers are needed for economic models (predicting when the bond market will decline and by how much) and traffic models (predicting when a car will reach an intersection or when a car will run a red light). Computer models that simulate molecular movement or the behavior of galaxies require vast quantities of random numbers.

Uniform Distribution

High-speed computers have produced both a need for random digits that could be available immediately and a method by which those digits could be produced via algorithmic generators. These generators, if reliable, promise that the digits have been thoroughly shuffled and each digit is equally likely to be drawn. Still, the notion that a machine and a programmed formula can create randomness makes some mathematicians very uncomfortable.

Uniform Distribution

From a theoretical perspective, this distribution is a key one in risk analysis; many Monte Carlo software algorithms use a sample from this distribution (between zero and one) to generate random samples from other distributions (by inversion of the cumulative form of the respective distribution).

On the other hand, there are only a few real-life processes that have this form of uncertainty. These could include for example: the position of a particular air molecule in a room, the point on a car tyre where the next puncture will occur, the number of seconds past the minute that the current time is, or the length of time that one may have to wait for a train. In oil exploration, the position of the oil-water contact in a potential prospect is also often considered to be uniformly continuously distributed.

Uniform Distribution

For the distribution to apply to each situation, implied assumptions need to hold, and it is the validity of these assumptions that can be questioned. In the example concerning the waiting time for a train, one would need to assume that trains arrive in regular intervals but that we have no knowledge of the current time, not of other indicators (sound, wind) that a train is in the process of arriving. For this reason, the distribution is sometimes called the "no knowledge" distribution.

Uniform Distribution

Example

Suppose that $X \sim U(0,1)$. Let $Y = g(X) = aX + b$, $a > 0$.

(a) Find the p.d.f. $f_Y(y)$ of Y.

(b) Calculate the value of $E(Y)$ and $Var(X)$.

Solution. (a) Since all possible values of X is between 0 and 1, and $y = g(x) = ax + b$, $(a > 0)$, the possible values taken by Y is between b and $a + b$. If $b \leqslant y \leqslant a + b$, then

$$f_Y(y) = f_X\left(\frac{y-b}{a}\right)\frac{1}{a} = \frac{1}{a}.$$

If $y < b$ or $y > a + b$, then $f_Y(y) = 0$. That means $Y \sim U(b, a + b)$.

Uniform Distribution

Example

Suppose that $X \sim U(0,1)$. Let $Y = g(X) = aX + b$, $a > 0$.

(a) Find the p.d.f. $f_Y(y)$ of Y.

(b) Calculate the value of $E(Y)$ and $Var(X)$.

Solution. (b)

$$E(Y) = E(aX + b) = aE(X) + b = a + b.$$

$$Var(Y) = Var(aX + b) = a^2 Var(X) = \frac{a^2}{12}.$$

$\square$

Uniform Distribution

Example

Suppose that $X \sim U(-1,1)$. Let $Y = g(X) = X^2$.
(a) Find the p.d.f. $f_Y(y)$ of Y.
(b) Calculate the value of $E(Y)$ and $Var(X)$.

Solution. (a) Since $X \sim U(-1,1)$, the p.d.f. of X is

$$f(x) = \begin{cases} \dfrac{1}{2} & \text{for } -1 \leqslant x \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

Because $y = g(x) = x^2$, the possible values taken by Y is between 0 and 1.

Uniform Distribution

Solution. If $0 < y < 1$, then

$$f_Y(y) = \frac{1}{2\sqrt{y}}\big[f_X(\sqrt{y}) + f_X(-\sqrt{y})\big] = \frac{1}{2\sqrt{y}}\big(\frac{1}{2} + \frac{1}{2}\big) = \frac{1}{2\sqrt{y}}.$$

If $y \leqslant 0$ or $y \geqslant 1$, then $f_Y(y) = 0$.

(b)

$$E(Y) = E(X^2) = \int_{-\infty}^{+\infty} x^2 f_X(x)dx = \int_{-1}^{1} x^2 \cdot \frac{1}{2}dx = \frac{1}{3}.$$

$$Var(Y) = E\big[(X^2)^2\big] - [E(X^2)]^2 = \int_{-1}^{1} x^4 \cdot \frac{1}{2}dx - \frac{1}{9} = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}.$$

$\square$

Section 2.5 Continuous Random Variables

- 2.5.1 Uniform Distribution
- 2.5.2 Exponential Distribution
- 2.5.3 Normal Distribution

Exponential Distribution

The exponential distribution is the continuous analogue of the geometric distribution, that is, it also does not deteriorate with time. By this we mean that if the lifetime of an item is exponentially distributed, then an item that has been in use for ten (or any number of) hours is as good as a new item in regards to the amount of time remaining until the item fails. In fact, the exponential distribution is the probability distribution that describes the time between events in a Poisson process, i.e., a process in which events occur continuously and independently at a constant average rate.

Exponential Distribution

Definition

Let X be a continuous random variable whose density function is of the form

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & for\ x > 0, \\ 0 & for\ x \leqslant 0, \end{cases}$$

where $\lambda > 0$ is the scale parameter. We say that X follows a ***exponential distribution*** *with parameter λ. We write that $X \sim Exp(\lambda)$. The case where $\lambda = 1$ is called the standard exponential distribution.*

The corresponding d.f. of X is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{for } x \geqslant 0, \\ 0 & \text{for } x < 0, \end{cases}$$

Exponential Distribution

We are sure about $f(x)$ is a p.d.f. since

$$\int_{-\infty}^{\infty} f(x)dx = \int_{0}^{+\infty} \lambda e^{-\lambda x} dx = 1.$$

The exponential distribution is usually used to model the time until something happens in the process.

Property

Suppose that X is a random variable which has exponential distribution with parameter λ. Then we have

$$E(X) = \frac{1}{\lambda} \quad \text{and} \quad Var(X) = \frac{1}{\lambda^2}. \tag{21}$$

Exponential Distribution

Solution. The mean of the exponential distribution is calculated using integration by parts as follows.

$$
\begin{aligned}
E(X) &= \int_0^{+\infty} \lambda x e^{-\lambda x} dx \\
&= \lambda\Big(\frac{-xe^{-\lambda x}}{\lambda}\Big|_0^{+\infty} + \frac{1}{\lambda}\int_0^{+\infty} e^{-\lambda x} dx\Big) \\
&= \lambda\Big(0 + \frac{1}{\lambda}\frac{-e^{-\lambda x}}{\lambda}\Big|_0^{+\infty}\Big) = \frac{1}{\lambda}.
\end{aligned}
$$

So one can see that as λ gets larger, the thing in the process we are waiting for to happen tends to happen more quickly, hence we think of λ as a rate.

Exponential Distribution

Solution. From the first and second moments we can compute the variance as

$$\begin{aligned} Var(X) =& E(X^2) - [E(X)]^2 \\ =& \int_0^{+\infty} x^2 e^{-\lambda x} dx - (1/\lambda)^2 \\ =& \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \end{aligned}$$

$\square$

Exponential Distribution

We can show that the exponential distribution, just as the geometric distribution, possesses the memoryless property. That is, if $X \sim Exp(\lambda)$, then

$$P(X > t + s | X > t) = P(X > s) \quad \text{for } s, t \geqslant 0.$$

Actually, only the geometric and exponential distributions possess this memoryless property. The geometric distribution describes the time until the first success. The exponential probability density function describes the time of the first event when events occur at a constant rate.

Exponential Distribution

Example

The lifetime (in years) of a radio has an exponential distribution with parameter $\lambda = 1/10$. If we buy a five-year-old radio, what is the probability that it will work for less than 10 additional years?

Solution. Let X be the total lifetime of the radio. We have that $X \sim Exp(\lambda = 1/10)$. We seek

$$\begin{aligned} P(X \leqslant 15 | X > 5) &= 1 - P(X > 15 | X > 5) \\ &= 1 - P(X > 10) = P(X \leqslant 10) \\ &= \int_0^{10} \frac{1}{10} e^{-x/10} dx = -e^{-x/10} \Big|_0^{10} \\ &= 1 - e^{-1} \approx 0.6321. \end{aligned}$$

$\square$

Exponential Distribution

Example

Jobs are sent to a printer at an average of 3 jobs per hour. (a) What is the expected time between jobs? (b) What is the probability that the next job is sent within 5 minutes?

Solution. Job arrivals represent rare events, thus the time T between them is exponential with rate 3 jobs/hour, i.e., $\lambda = 3$.
(a) Thus $E(T) = 1/\lambda = 1/3$ hours or 20 minutes.
(b) Using the same units (hours) we have 5 min.$= 1/12$ hours. Thus we compute

$$P(T < 1/12) = 1 - e^{-3 \cdot \frac{1}{12}} = 1 - e^{-\frac{1}{4}} = 0.2212.$$

$\square$

Exponential Distribution

Example

(Hits on a webpage) Suppose that we are told that, on average, there are 2 hits per minute on a specific web page. I start to observe this web page at a certain time point 0, and decide to model the waiting time till the first hit Y (in minutes) using an exponential distribution.

Solution. For modeling the distribution of Y using the exponential, we need an appropriate value for λ, the rate parameter. Since, on average there are 2 hits per minute, the average waiting time between hits of 0.5 minutes. That is we may use this value as the expected value for $Y : E(Y) = 0.5$.

Exponential Distribution

Solution. Since we know, that $E(Y)$ for an exponential RV is $1/\lambda$, setting $1/\lambda = 0.5$ we get $\lambda = 2$ as a reasonable choice for the rate parameter. If our model is correct, λ describes the rate, at which this web page is hit! Using the above model we can answer questions like...

What is the probability that we have to wait at most 40 seconds to observe the first hit?

(Since we are working in time units of minutes, we need to express the 40 seconds above as 2/3 minutes.) Thus, we compute the probability that we have to wait at most 2/3 min to observe the first hit:

$$P(Y \leqslant 2/3) = 1 - e^{-2 \cdot \frac{2}{3}} \approxeq 0.736.$$

Exponential Distribution

Solution. Note carefully that we just used the cdf of Y (that we know) for the above calculation.

How long do we have to wait at most, to observe a first hit with a probability of 0.9? This is the reverse of what we have computed so far, because here we want to find a t, for which $P(Y \leqslant t) = 0.9$:

$$\begin{aligned} P(Y \leqslant t) =& 0.9 \;\Leftrightarrow 1 - e^{-2t} \\ =& 0.9 \;\Leftrightarrow e^{-2t} = 0.1 \; t = -0.5 \ln 0.1 \approx 1.15 (min) \end{aligned}$$

that's approx. 69 seconds.
In the Hits on a web page example we said that we start to observe the web page a time point 0.

Exponential Distribution

Solution. Does the choice of this time point affect our analysis in any way?

If there is no hit after the first minute after we started, what is the probability, that we have to wait for another 40 seconds for the first hit?

The probability we want to compute is a conditional probability. The probability we need is
P(wait for 1 min and 40 sec for the first hit | no hits during the 1st min)
Recall that, we used the random variable Y to denote the waiting time till the first hit. It is easy to see that the required probability is then $P(Y \leqslant 5/3 | Y > 1)$
Now, recall that under the model we assumed, $Y \sim Exp(2.0)$

Exponential Distribution

Solution.

$$P(Y \leqslant 5/3 | Y > 1) = \frac{P(Y \leqslant 5/3 \cap Y > 1)}{P(Y > 1)}$$
$$= \frac{P(1 < Y \leqslant 5/3)}{1 - P(Y \leqslant 1)} = \frac{e^{-2} - e^{-10/3}}{e^{-2}} = 0.736.$$

This is exactly the same probability as when we started at time 0. $\square$

Exponential Distribution

Example

There is an equipment. Let $N(t)$ be the failure time of this equipment at any time length t. Assume that $N(t)$ has the Poisson distribution $P(\lambda t)$. Find the distribution of the interval time T between two failure time.

Solution. Since $N(t) \sim P(\lambda t)$,

$$P(N(t) = x) = e^{-\lambda t}\frac{(\lambda t)^x}{x!}, \qquad x = 0, 1, 2, \cdots$$

If $t > 0$, $\{T > t\} = \{N(t) = 0\}$, then

$$P(T > t) = P(N(t) = 0) = e^{-\lambda t}.$$

So

$$P(T \leqslant t) = \begin{cases} 1 - \exp(-\lambda t) & \text{for } t > 0, \\ 0 & \text{for } t \leqslant 0. \end{cases}$$

That means $T \sim Exp(\lambda)$. $\square$

Exponential Distribution

Example

Let $X \sim U(0,1)$ and $Y = -\theta lnX$, where $\theta > 0$. Find the p.d.f. $f_Y(y)$ of Y.

Solution. First, note that the possible values of the random variable Y are those located in the interval $(0, \infty)$. Next, we have that $g^{-1}(y) = e^{-y/\theta}$, so that

$$f_Y(y) = f_X(e^{-y/\theta})\left|\frac{d}{dz}e^{-y/\theta}\right| = 1 \cdot \left| -\frac{1}{\theta}e^{-y/\theta}\right| = \frac{1}{\theta}e^{-y/\theta}$$

for $0 < y < \infty$. Thus, we can assert that Y has an exponential distribution with parameter $1/\theta$. $\square$

Exponential Distribution

We make use of this result in simulation. Indeed, if we wish to generate an observation of a random variable following an exponential distribution with parameter $\lambda = 2$ (for instance), it suffices to generate an observation x of a $U(0,1)$ distribution, and then to apply the transformation $y = -(1/2)\ln x$. Therefore, it is not necessary to write a special computer program to generate observations of exponential distributions. Computer languages in general, and even many pocket calculators, allow us to generate pseudo-random observations of uniform distributions.

Exponential Distribution

In queuing theory, the service times of agents in a system (e.g. how long it takes for a bank teller etc. to serve a customer) are often modeled as exponentially distributed variables. (The arrival of customers for instance is also modeled by the Poisson distribution if the arrivals are independent and distributed identically.)

But in real-world scenarios, the assumption of a constant rate (or probability per unit time) is rarely satisfied. For example, the rate of incoming phone calls differs according to the time of day. But if we focus on a time interval during which the rate is roughly constant, such as from 2 to 4 p.m. during work days, the exponential distribution can be used as a good approximate model for the time until the next phone call arrives.

Exponential Distribution

Reliability theory and reliability engineering also make extensive use of the exponential distribution. Because of the memoryless property of this distribution, it is well-suited to model the constant hazard rate portion of the bathtub curve used in reliability theory. It is also very convenient because it is so easy to add failure rates in a reliability model.

The exponential distribution is however not appropriate to model the overall lifetime of organisms or technical devices, because the "failure rates" here are not constant: more failures occur for very young and for very old systems.

Section 2.5 Continuous Random Variables

- 2.5.1 Uniform Distribution
- 2.5.2 Exponential Distribution
- 2.5.3 Normal Distribution

Normal Distribution

The normal curve was developed mathematically in 1733 by De Moivre as an approximation to the binomial distribution. His paper was not discovered until 1924 by Karl Pearson. Laplace used the normal curve in 1783 to describe the distribution of errors. Subsequently, Gauss used the normal curve to analyze astronomical data in 1809. The normal curve is often called the Gaussian distribution.

Normal Distribution

Many numerical populations have distributions that can be fit very closely by an appropriate normal curve. Examples include heights, weights, blood pressure and other physical characteristics, lengths of objects produced by machines, measurement errors in scientific experiments, anthropometric measurements on fossils, reaction times in psychological experiments, measurements of intelligence and aptitude, scores on various tests, and numerous economic measures and indicators. Even when the underlying distribution is discrete, the normal curve often gives an excellent approximation.

Normal Distribution

Definition

Let X be a continuous random variable that can take any real value. If its density function is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \quad for -\infty < x < \infty, \tag{22}$$

then we say that X has a ***normal (or Gaussian) distribution*** *with parameters μ and σ^2, where $\mu \in \mathbb{R}$ and $\sigma > 0$. We write that $X \sim N(\mu, \sigma^2)$.*

Normal Distribution

If $X \sim N(\mu, \sigma^2)$, then the d.f. of X is

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt.$$

We can verify that the nonnegative function defined in equation (22) is a valid probability density function by showing that

$$\int_{-\infty}^{\infty} f(x)dx = 1. \tag{23}$$

If we let $y = (x - \mu)/\sigma$, then

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{1}{2}y^2\Big)dy.$$

Normal Distribution

We shall now let

$$I = \int_{-\infty}^{\infty} \exp\Big(-\frac{1}{2}y^2\Big)dy.$$

It follows that

$$\begin{aligned} I^2 =& I \cdot I = \int_{-\infty}^{\infty} \exp\Big(-\frac{1}{2}y^2\Big)dy \int_{-\infty}^{\infty} \exp\Big(-\frac{1}{2}z^2\Big)dz \\ =& \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\Big[-\frac{1}{2}(y^2+z^2)\Big]dydz. \end{aligned}$$

We shall now change the variables in this integral from y and z to the polar coordinate r and θ by letting $y = r\cos\theta$ and $z = r\sin\theta$. Then, since $y^2 + z^2 = r^2$,

$$I^2 = \int_0^{2\pi}\int_0^{\infty} \exp\Big(-\frac{1}{2}r^2\Big)rdrd\theta = 2\pi.$$

Therefore, $I = (2\pi)^{1/2}$ and the correctness of (23) has been established.

Normal Distribution

Property

Suppose that $X \sim N(\mu, \sigma^2)$. Then we have

$$E(X) = \mu \quad \text{and} \quad Var(X) = \sigma^2. \tag{24}$$

Proof.

$$\begin{aligned}
E(X) &= \int_{-\infty}^{+\infty} x f(x) dx = \int_{-\infty}^{+\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx \\
&\overset{\frac{x-\mu}{\sigma}=z}{=} \int_{-\infty}^{+\infty} (\mu + \sigma z) \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2} \sigma dz \\
&= \mu + \sigma \int_{-\infty}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\
&= \mu + \sigma \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \, d\frac{z^2}{2} = \mu,
\end{aligned}$$

Normal Distribution

Proof.

$$
\begin{aligned}
Var(X) &= \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx \\
&= \int_{-\infty}^{+\infty} (x-\mu)^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx \\
&= -\sigma^2 \cdot (x-\mu) \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \Big|_{-\infty}^{+\infty} \\
&\qquad + \sigma^2 \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx \\
&= \sigma^2.
\end{aligned}
$$

$\square$

Normal Distribution

Observe the p.d.f. $f(x)$.

All normal distributions have the same general shape, namely that of a bell. So the Gaussian distribution is sometimes informally called the bell curve. The functions $f(x)$ are symmetrical with respect to the expected value μ. That is, $f(x-\mu) = f(x+\mu)$ for all $x \in R$. The points $\mu - \sigma$ and $\mu + \sigma$ are those where the function f changes its direction of concavity. And

$$f_{\max} = f(\mu) = \frac{1}{\sqrt{2\pi}\sigma}.$$

Normal Distribution

The parameter μ is called positional parameter in normal distribution, and σ^2 is called shape parameter. The larger σ is, the more flattened the curve is. Conversely, if σ is small, then the curve is concentrated around the mean μ.

The normal distribution with parameter values $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution. A random variable having a ***standard normal distribution*** is called a standard normal random variable and will be denoted by Z. The p.d.f. of Z is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad \text{for } -\infty < z < \infty, \tag{25}$$

The graph of $\phi(z)$ is called the standard normal (or z) curve. The d.f. of Z is

$$\Phi(z) := P(Z \leqslant z) = \int_{-\infty}^{z} \phi(y) dy.$$

Normal Distribution

By Observing the functions $\phi(z)$ and Φ, we can find that

Property

(i) $\Phi(x) + \Phi(-x) = 1$, (ii) $\Phi(0) = 1/2$.

The importance of standard normal distribution is that it is usually regarded as a reference distribution from which information about other normal distributions can be obtained.

In fact, we have

Normal Distribution

Property

If $X \sim N(\mu, \sigma^2)$, *then the d.f.* $F(x)$ *of* X *is given by* $\Phi\Big(\dfrac{x-\mu}{\sigma}\Big)$, *i.e.,*

$$F(x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big).$$

Proof. Let $s = \frac{t-\mu}{\sigma}$.

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,dt = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{s^2}{2}}\,ds = \Phi\Big(\frac{x-\mu}{\sigma}\Big).$$

$\square$

Normal Distribution

Property

If X has a normal distribution with mean μ and standard deviation σ, then

$$Z = g(X) = \frac{X-\mu}{\sigma}$$

has a standard normal distribution. Thus

$$\begin{aligned} P(a \leqslant X \leqslant b) =& P(\frac{b-\mu}{\sigma} \leqslant Z \leqslant \frac{a-\mu}{\sigma}) \\ =& \Phi\Big(\frac{b-\mu}{\sigma}\Big) - \Phi\Big(\frac{a-\mu}{\sigma}\Big), \end{aligned}$$

$$P(X \leqslant a) = \Phi(\frac{a-\mu}{\sigma}), \qquad P(X \geqslant b) = 1 - \Phi(\frac{b-\mu}{\sigma}).$$

Normal Distribution

Proof. Indeed, we have

$$z = g(x) = (x - \mu)/\sigma \Leftrightarrow g^{-1}(z) = \mu + \sigma z.$$

Since $\sigma > 0$,

$$f_Z(z) = f_X(\mu + \sigma z)\left|\frac{d}{dz}\mu + \sigma z\right| = \frac{1}{\sqrt{2\pi}\sigma}\exp(-z^2/2)|\sigma| = \phi(z)$$

for $-\infty < z < \infty$. $\square$

Normal Distribution

Corralary

Suppose that $X \sim N(\mu, \sigma^2)$. *Let* $Y = aX + b$. *Then* Y *has a normal distribution with parameters* $a\mu + b$ *and* $a^2\sigma^2$.

Table of Normal Probabilities in Appendix gives $\Phi(z) = P(Z \leqslant z)$, the area under the standard normal density curve to the left of z, for $z = 0, 0.01, \cdots, 3.98, 3.99$. From this table, various other probabilities involving Z can be calculated. By Proposition 17 and Proposition 18, this table can also be used to compute probabilities for any other values of μ and σ under consideration.

Normal Distribution

Example

Let us determine the following standard normal probabilities: (a)$P(Z \leqslant 1.25)$, (b) $P(Z > 1.25)$, (c) $P(Z \leqslant -1.25)$, (d) $P(-0.38 \leqslant Z \leqslant 1.25)$.

Solution. (a) $P(Z \leqslant 1.25) = \Phi(1.25)$, a probability that is tabulated in Appendix Table of Normal Probabilities at the intersection of the row marked 1.2 and the column marked 0.05. The number there is 0.89435, so $P(Z \leqslant 1.25) = 0.89435$.

(b) $P(Z > 1.25) = 1 - P(Z \leqslant 1.25) = 1 - \Phi(1.25)$. Then $\Phi(1.25) = 0.89435$ implies that $P(Z > 1.25) = 0.10565$. Since Z is a continuous r.v., $P(Z \geqslant 1.25) = 0.10565$.

Normal Distribution

Solution. (c) $P(Z \leqslant -1.25) = \Phi(-1.25)$, a lower-tail area. By Proposition 16, $\Phi(-1.25) = 0.10565$. By symmetry of the z curve, this is the same answer as in part (b).

(d) From Section 2.2, if X is a continuous r.v. with d.f. $F(x)$, then $P(a \leqslant X \leqslant b) = F(b) - F(a)$. Thus

$$\begin{aligned} P(-0.38 \leqslant Z \leqslant 1.25) &= \Phi(1.25) - \Phi(-0.38) \\ &= \Phi(1.25) - (1 - \Phi(0.38)) \\ &= 0.89435 - 0.35197 = 0.54238. \end{aligned}$$

$\square$

Normal Distribution

Example

The time that it takes a driver to react to the brake lights on a decelerating vehicle is critical in helping to avoid rear-end collisions. Suppose that reaction time for an in-traffic response to a brake signal from standard brake lights can be modeled with a normal distribution having mean value 1.25sec and standard deviation of 0.46sec. What is the probability that reaction time is between 1.00sec and 1.75sec?

Normal Distribution

Solution. If we let X denote reaction time, then standardizing gives

$$1.00 \leqslant X \leqslant 1.75$$

if and only if

$$\frac{1-1.25}{0.46} \leqslant \frac{X-1.25}{0.46} \leqslant \frac{1.75-1.25}{0.46}.$$

Thus

$$\begin{aligned}
&P(1 \leqslant X \leqslant 1.75) \\
&= P\Big(\frac{1-1.25}{0.46} \leqslant Z \leqslant \frac{1.75-1.25}{0.46}\Big) \\
&= P(-0.54 \leqslant Z \leqslant 1.09) = \Phi(1.09) - (1-\Phi(0.54)) \\
&= 0.86214 - (1-0.70540) = 0.56754.
\end{aligned}$$

$\square$

Normal Distribution

Similarly, if we view 2sec as a critically long reaction time, the probability that actual reaction time will exceed this value is

$$P(X > 2) = P\Big(Z > \frac{2 - 1.25}{0.46}\Big)$$

$$= P(Z > 1.63) = 1 - \Phi(1.63) = 0.05155.$$

Normal Distribution

Example

Suppose that $X \sim N(3, 4)$. Let $Y = 2X + 1$.

(a) Find the value of $P(2 < X < 3)$.

(b) If we let $Y = aX + 4$, then find the value a such that $P(Y \leqslant 7) = 1/2$.

Solution. Let $F(x)$ be the d.f. of X.
(a)

$$\begin{aligned} P(3 < X < 4) &= F(4) - F(3) \\ &= \Phi\Big(\frac{4-3}{2}\Big) - \Phi\Big(\frac{3-3}{2}\Big) \\ &= 0.69146 - 0.5 = 0.19146. \end{aligned}$$

Normal Distribution

Solution. (b) Since $F(E(Y)) = 0.5$ and $P(Y \leqslant 7) = 1/2$, $E(Y) = 7$. We have

$$7 = E(Y) = E(aX + 4) = aE(X) + 4 = 3a + 4.$$

Thus $a = 1$. $\square$

Normal Distribution

Example

The breakdown voltage of a randomly chosen diode of a particular type is known to be normally distributed. What is the probability that a diode's breakdown voltage is within 1 standard deviation(SD) of its mean value?

Solution. This question can be answered without knowing either μ or σ, as long as the distribution is known to be normal; the answer is the same for any normal distribution:

$$
\begin{aligned}
& P(X \text{ is within 1 standard deviation of its mean}) \\
&= P(|X-\mu| \leqslant \sigma) = \ P(\mu-\sigma \leqslant X \leqslant \mu+\sigma) \\
&= P\Big(\frac{\mu-\sigma-\mu}{\sigma} \leqslant Z \leqslant \frac{\mu+\sigma-\mu}{\sigma}\Big) \\
&= P(-1 \leqslant Z \leqslant 1) \\
&= \Phi(1) - \Phi(-1) = 0.6826.
\end{aligned}
$$

$\square$

Normal Distribution

Similarly, the probability that X is within 2 standard deviations is

$$P(|X - \mu| \leqslant 2\sigma) = P(-2 \leqslant Z \leqslant 2) = 0.9544,$$

and within 3 standard deviations is

$$P(|X - \mu| \leqslant 3\sigma) = P(-3 \leqslant Z \leqslant 3) = 0.9974.$$

Normal Distribution

Standardizing amounts to nothing more than calculating a distance from the mean value and then reexpressing the distance as some number of standard deviations. Thus, if $\mu = 100$ and $\sigma = 15$, then $x = 130$ corresponds to $z = (130 - 100)/15 = 30/15 = 2.00$. That is, 130 is 2 standard deviations above (to the right of) the mean value. Similarly, standardizing 85 gives $(85 - 100)/15 = 1.00$, so 85 is 1 standard deviation below the mean.

Normal Distribution

If the population distribution of a variable is (approximately) normal, then

(i) Roughly 68% of the values are within 1 SD of the mean,

(ii) Roughly 95% of the values are within 2 SDs of the mean,

(iii) Roughly 99.7% of the values are within 3 SDs of the mean.

That means it is unusual to observe a value from a normal population that is much farther than 2 standard deviations from μ. These results will be important in the development of hypothesis-testing procedures in statistics.

Normal Distribution

The normal distribution sometimes can be used to approximate both the binomial and Poisson distributions. A common rule-of-thumb is to approximate the binomial whenever n (the number of trials) is larger than 30. If $np < 5$, then use the Poisson for the approximation with $\lambda = np$. If $np \geqslant 5$, then use the normal for the approximation with $\mu = np$ and $\sigma^2 = np(1-p)$.

Normal Distribution

Furthermore, the normal can be used to approximate the Poisson whenever $\lambda > 30$. When using a continuous distribution (like the normal) to approximate a discrete distribution (like the Poisson or binomial), the interval between the discrete values is usually split halfway.

For example, if we desire to approximate the probability that a Poisson random variable will take on the values 29, 30, or 31 with a continuous distribution, then we would determine the probability that the continuous random variable is between 28.5 and 31.5.

Normal Distribution

Example

A software company has received complaints regarding their responsiveness for customer service. They have decided to analyze the arrival pattern of phone calls to customer service and have determined that the arrival of complaints in any one hour is a Poisson random variable with a mean of 120. The company would like to know the probability that in any one hour 140 or more calls arrive.

Solution. To determine that probability, let N be a Poisson random variable with $\lambda = 120$, let X be a random variable with $\mu = \sigma^2 = 120$ and let Z be a standard normal random variable. The above question is answered as follows:

$$\begin{aligned} P(N \geqslant 140) =& P(X > 139.5) = P(Z > (139.5 - 120)/10.95) \\ =& P(Z > 1.78) = 1 - \Phi(1.78) = 1 - 0.9625 = 0.0375. \end{aligned}$$

$\square$

Normal Distribution

Example

A manufacturing process produces 10% defective items. A sample of 200 items is drawn at random. Let X be the number of defective items in the sample. Use a normal distribution to calculate (approximately) the probability $P(X = 20)$.

Solution. We may assume (see the first remark below) that the random variable X has a binomial distribution with parameters $n = 200$ and $p = 0.1$. So we have that $\mu = np = 200 \cdot 0.1 = 20 > 5$. We set

$$\begin{aligned} P(X = 20) &\approx f_Y(20) \quad \text{where } Y \sim N(20, 18) \\ &= \frac{1}{\sqrt{2\pi}\sqrt{18}} \exp\{-(20-20)^2/(2 \cdot 18)\} \approx 0.0904. \end{aligned}$$

$\square$

Normal Distribution

We can also proceed as follows:

$$
\begin{aligned}
P(X = 20) &= P(20 \leqslant X \leqslant 20) \\
&\approx P\Big(Z \leqslant \frac{20 + 0.5 - 20}{\sqrt{18}}\Big) - P\Big(Z \leqslant \frac{20 - 0.5 - 20}{\sqrt{18}}\Big) \\
&\approx \Phi(0.12) - \Phi(-0.12) = 2 \cdot 0.5478 - 1 = 0.0956.
\end{aligned}
$$

Normal Distribution

Remark

(i) *In the statement of the problem, it is not specified whether the sample is taken with or without replacement. Since the value of the size N of the lot, which is not specified either, we must assume that all the items in the sample have a 0.1 probability of being defective, independently from one item to the other.*

(ii) *The exact value, obtained by using the binomial distribution, is*

$$P(X = 20) = \binom{200}{20} 0.1^{20} 0.9^{180} \approx 0.0936.$$

Normal Distribution

Remark

(iii) *The Poisson approximation should work well in this example, because n is very large and p is relatively small. We find that*

$$P(X = 20) = e^{-20}\frac{20^{20}}{20!} \approx 0.0888.$$

Thus, the normal approximation is actually better in this particular example.

The end

Thank you for your patience !